

Byzantine-Fault-Tolerant Hierarchical Navigation for GNSS-Denied UAV Swarms: Architecture, Theoretical Analysis, and Empirical Validation

Oleksandr Kalynovskyi Independent Researcher

Email: kalinovskyi.research@proton.me

ORCID: 0009-0009-1437-3252

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Abstract—Cooperative navigation of unmanned aerial vehicle (UAV) swarms under Global Navigation Satellite System (GNSS) denial is increasingly challenged by adversarial peers; defence at the state-estimation layer is largely absent. We present and empirically validate, in simulation (Gazebo/PX4 SITL together with a protocol-layer Python harness), a flight-targeted architecture simultaneously addressing six concerns: Byzantine-fault-tolerant consensus over navigation data, a hierarchical anchor/follower topology, reputation-weighted multi-hop position relay, automatic failover, ultra-wideband cross-verification of reported positions, and capacity-aware load balancing under progressive attrition. The core primitive is a *Visual Proof of Observation*: cryptographically signed hashes of terrain feature descriptors that neighbouring UAVs verify by physical revisit. Reputation is updated asymmetrically and mapped to factor-graph noise covariances. We derive a Byzantine safety bound, a closed-form parametric estimate of the naive-Byzantine exclusion time, and a per-hop error propagation model, and execute a seven-phase empirical campaign. Key results: Proof-of-Observation AUC of 0.9996, zero false exclusions across 30 trials, circular error probable of 23 m at three relay hops, 99th-percentile failover of 6.24 s, spoofing detection rate of 98.3% with zero false positives and 99th-percentile detection time of 600 ms, and zero over-capacity interval under attrition with admission control. A vanilla Raft baseline quantifies the Byzantine-versus-crash-fault-tolerance capability gap: under a single-Byzantine leader-claimer attack, Raft is compromised indefinitely while our protocol excludes the adversary within one second.

I. INTRODUCTION

UNMANNED aerial vehicle (UAV) swarms are increasingly deployed in contested environments where GNSS signals are unavailable, degraded, or deliberately denied. Recent operational experience in military and civil domains has underscored two coupled vulnerabilities: GNSS-dependent navigation collapses under electronic warfare, and cooperative navigation schemes that share position information between

vehicles assume that every participant reports honestly. A single compromised UAV—captured, sensor-faulted, or infected with malicious firmware—that transmits falsified odometry or absolute-position estimates can corrupt the navigation solution of the entire swarm without detection.

Three research streams have independently matured in the past five years. First, cooperative GNSS-denied navigation has produced flight-targeted systems such as the Distributed PNT Chain [1], Omni-swarm [2], Kimera-Multi [3], and Swarm-SLAM [4], and anchor-free multi-sensor fusion schemes such as DTVIRM-Swarm [5] have added magnetic and monocular cues. Second, Byzantine-fault-tolerant (BFT) protocols for robot swarms have evolved from abstract consensus [6] to deployed token economies [7] and blockchain-secured cluster hierarchies [8]. Third, reputation and trust frameworks for multi-agent systems have converged on asymmetric update rules and physical-layer observations [9]–[11]. Yet these streams have not merged at the flight-control layer. The two 2025 systems closest to a unified architecture—SwarmRaft [12] and the Byzantine Distributed GPS Detection (BDGD) algorithm [13]—each cover at most three of the design dimensions required for a deployable solution; an earlier and largely complementary system, the Byzantine Follow-The-Leader (BFTL) formalism of Castelló Ferrer *et al.* [14], covers four, but operates on discretised maps rather than a continuous flight-targeted state estimator.

This paper introduces a navigation architecture that simultaneously addresses six pillars: (i) BFT consensus over shared state, (ii) a hierarchical anchor/follower topology with roles assigned dynamically in flight, (iii) reputation-weighted multi-hop position relay, (iv) automatic failover within 10 seconds of anchor loss, (v) GNSS-spoofing detection via UWB cross-verification, and (vi) load balancing across anchors under progressive attrition. Based on a systematic review of approximately 90 publications spanning 2020–2026 across Byzantine-fault-tolerant swarms, GNSS-denied cooperative navigation, multi-robot reputation systems, hierarchical swarm control, GNSS-spoofing detection, and multi-hop position relay (see Section II footnote for review methodology), we find, to our knowledge, no published system integrating more than four of these pillars, and none doing so within a continuous-state flight-targeted factor-graph estimator. The architecture rests on a new primitive, the *Visual Proof of Observation* (PoO): each terrain-relative navigation (TRN) fix produces a crypto-

The methods described in Sections IV-B through IV-G are the subject of pending patent applications filed with the German Patent and Trademark Office (DPMA), Aktenzeichen 10 2026 001 712.2 and a related application. This manuscript describes the technical content of those filings for the purposes of academic disclosure. Empirical validation reported in Section VI was carried out in a two-tier framework (Gazebo/ROS2 for in-flight validation; a Python harness for protocol-layer measurement at scale) released as open-source companion code (Section XI-E).

graphically signed record containing a hash of terrain feature descriptors, which neighbouring UAVs verify by comparing their own descriptors when they later fly over the claimed region. Because feature descriptors are physically bound to terrain, illumination, and the learned extractor network, they cannot be fabricated by a vehicle that has not actually observed the location, distinguishing this mechanism from signal-level or purely statistical trust.

Our contributions are as follows. *First*, we formulate a visual PoO mechanism that gives distributed navigation a terrain-anchored verification primitive (physically bound to terrain, illumination, and the learned feature extractor; hard to forge under the threat model of Section III; the property holds in the practical sense given the threat model’s assumptions, not under the formal cryptographic UF-CMA definition), and we show how to integrate signed observations into an incremental smoothing and mapping (iSAM2) factor graph as reputation-weighted priors alongside UWB ranging factors. *Second*, we define an asymmetric reputation rule ($\alpha_{\text{neg}} \geq 2\alpha_{\text{pos}}$, preferably 3:1) that excludes Byzantine participants faster than honest peers accumulate trust while tolerating transient verification failures from environmental degradation. *Third*, we introduce a dynamic anchor-score assignment combining reputation, TRN confidence, imagery coverage, and load, together with a “request–offer–select” failover protocol with bounded recovery time. *Fourth*, we derive a safety condition for quorum-based exclusion, expose its limits when the quorum parameter M is fixed, give a closed-form parametric estimate for the expected naive-Byzantine exclusion time (under an inter-verifier-overlap assumption $\rho \leq 2$ characterised in Section V), and present a closed-form per-hop error-growth model. *Fifth*, we report a seven-phase empirical validation campaign of the proposed architecture in a Gazebo/ROS2 + Python harness: Proof-of-Observation AUC=0.9996 (FAR= 0%/FRR= 4.60% at $T_{\text{verify}} = 0.30$), 30/30 reputation-driven exclusion trials with zero false exclusions of honest peers, CEP=23 m at three relay hops ($> 4\times$ inside the 100-m operational ISR target) with empirical per-hop multiplier $\delta=0.088$ tightening the conservative $\delta = 0.15$ prior assumption, p99 anchor failover at 6.24 s across 1200 measured switches, 98.3% GNSS-spoofing detection with zero false positives at p99 detection time 0.60 s, and zero over-capacity interval under attrition with capacity-aware admission control. *Sixth*, we report a vanilla-Raft baseline comparison and a single-Byzantine adversarial measurement that exposes the BFT-vs-CFT capability gap.

The remainder of the paper is organised as follows. Section II surveys related work. Section III formalises the system and threat model. Section IV details the proposed approach in seven subsections. Section V presents the theoretical analysis. Section VI reports the empirical validation campaign across seven phases (Proof-of-Observation, reputation-driven exclusion, circular error probable versus relay depth, failover timing, GNSS-spoofing reaction, progressive attrition with admission control, and single-pillar ablation). Section VII presents a vanilla-Raft baseline comparison, including a Byzantine adversarial measurement that exposes the BFT-vs-CFT capability gap. Sections VIII–X report complementary abstract Monte Carlo validation of the reputation mechanism (asymmetry-ratio

latency, convergence-bound tightness, and Byzantine-fraction resilience). Section XI discusses limitations and deployment considerations. Section XII concludes.

II. RELATED WORK

We organise prior art into five domains and summarise design-space coverage of the most relevant systems in Table I.¹

A. Byzantine Fault Tolerance in UAV Swarms

The Byzantine fault model originates with Lamport, Shostak, and Pease [15], who established the foundational $n \geq 3f + 1$ requirement for consensus among n peers in the presence of f arbitrarily-faulty participants. Practical implementations have evolved through two main families described below.

Raft-family adaptations. The original Raft consensus algorithm [16] provides crash-fault tolerance with leader-based replication; UAV-specific extensions add Byzantine resilience on top of this substrate. SwarmRaft [12] performs median-based correction of spoofed GNSS positions among leader-collected peer reports and includes a Byzantine-resilient evaluation step on top of a crash-fault Raft core. Its design choice is not presented as a limitation by the authors, but the system lacks persistent reputation across rounds, weights all peers equally within a round, and retains a single elected leader. Zhang *et al.* [17] overlay Proof-of-Authority on Raft for UAV routing, not localisation. Zuo *et al.* [18] provide Raft-inspired leader election with explicit follower-loss failover but no reputation.

PBFT variants. Practical Byzantine Fault Tolerance, originally introduced by Castro and Liskov [19], has spawned a family of UAV-specific extensions. LAP-BFT [20] provides lightweight asynchronous BFT with vector commitments over 50 UAVs; DTPBFT [21] extends PBFT with a dynamic trust mechanism for UAV swarms; ACBFT [22] is an adaptive chained Byzantine-fault-tolerant consensus protocol for UAV ad-hoc networks; DPoS-PBFT [23] hybridises delegated proof-of-stake with PBFT finality. The concurrent reputation-enhanced PBFT variant RePA [24] adds trust scoring to defend against node-capture attacks at the network layer. Concurrent work on parallelised BFT for swarm robots (PTEE-BFT [25]) introduces a parallel-execution Byzantine consensus protocol

¹Review methodology: candidate publications were identified by keyword search in Google Scholar, arXiv, IEEE Xplore, and ACM Digital Library between February and April 2026 using combinations of $\{\textit{Byzantine}, \textit{BFT}, \textit{fault-tolerant}, \textit{reputation}, \textit{trust}\} \times \{\textit{UAV}, \textit{drone}, \textit{swarm}, \textit{multi-robot}\} \times \{\textit{navigation}, \textit{localisation}, \textit{SLAM}, \textit{GNSS}, \textit{GPS}\}$; inclusion criteria were peer-reviewed conference or journal venue (or arXiv preprint with an identifiable author affiliation), publication date 2020–2026, and at least partial coverage of one of the six pillars enumerated in Section I. The 2020 lower bound captures the post-2019 increase in UAV-swarm Byzantine-resilience publications; pre-2020 prior art consists primarily of generic distributed-systems BFT work whose UAV-specific instantiations appear only later. Of approximately 90 candidate publications screened, the systems explicitly compared in Table I are those that cover three or more pillars. The “six pillars” refer to design-space capabilities required for a deployable BFT-resilient swarm-navigation system; not all six are themselves BFT primitives (Pillar 6, capacity-aware load balancing under progressive attrition, is a topology-stability concern that becomes relevant under Byzantine-induced anchor loss rather than a BFT mechanism per se).

for blockchain-secured swarms, addressing throughput rather than navigation. None of these targets localisation. The recent COUAVChain [26], an RB-HotStuff-based variant building on the HotStuff BFT protocol [27], couples dynamic reputation, Bayesian inference, and hierarchical sharding for denied environments—architecturally closest to ours on the consensus side—but does not close the loop with spoofing detection or multi-hop position relay.

Blockchain-based swarm reputation. The IRIDIA/ULB line established Byzantine resilience for physical robot swarms [6], [7], [28], [29]. Strobel *et al.* [7] demonstrated a token economy on Pi-puck swarms that excludes adversaries autonomously. Castelló Ferrer *et al.* [30] extended the framework with secure multi-party computation for confidential missions. Among these, the BFTL system [14] is the closest prior navigation-aware work: it formalises the Byzantine Follow-The-Leader problem, publishes cryptographically weighted routes to a blockchain, and proves completion bounds under Byzantine leaders and followers (including a form of leader-failure tolerance arising from the completion guarantees). BFTL covers BFT, hierarchy, reputation, and position relay but operates over discretised graph-structured environments (with routes committed to blockchain) rather than a continuous probabilistic factor-graph state estimator. The blockchain settlement layer used in the published BFTL demonstration has route-commitment finality on the order of seconds per round [6] — a property of the chosen settlement layer rather than of the BFTL formalism itself, but the published instantiation as deployed is consequently incompatible with 50-ms control cycles; a future BFTL implementation over a faster settlement layer could in principle close this gap, and the BFT design choices that distinguish our work from BFTL are independent of the settlement-layer latency question (continuous-state factor-graph estimation, dynamic failover, and load balancing in particular). Alsamhi *et al.* [8] link blockchain trust to cluster-head hierarchy with energy awareness but without multi-hop relay or spoofing defence. Jensen *et al.* [31] and De Campos *et al.* [32] explored earlier blockchain-UAV integrations.

Distributed resilient estimation. The W-MSR track originating with LeBlanc *et al.* [33], with later extensions to trusted-node settings by Mitra *et al.* [34], provides formal scalar-state guarantees but lacks flight-targeted sensor-fusion detail.

B. GNSS-Denied Cooperative Navigation

The Distributed PNT Chain [1] from GTRI factors a centralised Kalman filter over swarm poses into an undirected acyclic Bayes graph, matching centralised accuracy at $\sim 22\%$ of the compute cost. It does not defend against peers that falsify odometry—a spoofed node propagates unchecked through the factor graph. Omni-swarm [2] and its successor D²SLAM [35] are canonical decentralised UWB+VIO systems; CCM-SLAM [36], COVINS [37], Kimera-Multi [3], and Swarm-SLAM [4] are the leading collaborative VIO/SLAM frameworks, evaluated on swarms of fewer than 20 agents. None defends against falsified loop closures or odometry;

statistical outlier rejection (GNC, robust kernels) is routine, but deliberate adversarial behaviour is out of scope. Moroncelli *et al.* [38] extended Swarm-SLAM with blockchain-based geometric Byzantine detection—the closest prior coupling of reputation with SLAM, although operating on ground robots and without hierarchical failover.

The “Collaborative Integrated Navigation” scheme of Zhang *et al.* [39] models uncertainty propagation from a leader to followers via UWB ranging but employs only two hierarchy levels with a statically assigned leader, no reputation, and no failover. The air-ground scheme of Yue *et al.* [40] uses high-altitude UAVs as fixed-role mobile UWB anchors without dynamic election, reputation propagation, or failover. DTVIRM-Swarm [5] is the most comprehensive honest-peer anchor-free fusion to date (MIMU + magnetic + monocular + UWB with χ^2 outlier filtering), but assumes statistical rather than adversarial outliers. Causa *et al.* [41] present a decentralised cooperative navigation solution for a UAV swarm in GNSS-degraded conditions using a rotating coordinator architecture that collects cross-covariance terms to match centralised accuracy while remaining resilient to individual node failures; unlike the proposed system, Byzantine-peer detection and automatic anchor failover are not addressed.

C. Reputation Systems for Multi-Robot Systems

Yemini *et al.* [9] injected stochastic physical-layer trust into consensus, breaking the classical $< 50\%$ malicious bound. Ballotta *et al.* [11] added time-decaying confidence for tunable convergence. Mallmann-Trenn, Cavorsi, and Gil [10] extended this idea to flocking through “crowd vetting” of trust opinions. FLTrust [42] is the federated-learning analogue. Closer to the adversarial cooperative-localisation setting, Fang *et al.* [43] present an information-theoretic analysis of trustworthy UAV cooperative localisation against malicious peers; their treatment is single-hop and does not propose a multi-hop reputation-weighted noise-covariance propagation model, which remains the distinguishing mechanism in our architecture. Most of this work operates at the decision or aggregation layer, not the flight-control estimator.

D. GNSS-Spoofing Detection in Swarms

GPS spoofing of civilian UAVs was first demonstrated experimentally by Kerns *et al.* [44]. Mykytyn *et al.* [45] subsequently introduced UWB cross-verification against the Spherical Law of Cosines for GPS-spoofing detection in UAV swarms, demonstrated for small-swarm settings on consumer-grade hardware; their evaluation does not extend to swarm-wide multi-hop relay or hierarchical anchor/follower failover. Single-UAV anti-spoofing schemes such as Chimera authentication [46] and SemperFi [47] address signal-level integrity at one receiver but do not extend to swarm-wide cross-verification.

E. Hierarchical Swarms, Position Relay, and Load Balancing

The formation-control hierarchy of Chen *et al.* [48] scales but does not consider adversarial peers. Multi-hop cooperative

localisation has been studied extensively in the honest-peer setting [49], [50]. For benign (non-adversarial) localization dropout, Horyna *et al.* [51] show that UAV swarms adapt better to anchor localization loss than single robots by leveraging relative inter-agent information; this is complementary to the proposed system, which addresses adversarial Byzantine anchor behavior rather than benign sensor dropout, but to the best of our knowledge no published analysis addresses adversarial multi-hop UWB error propagation; existing multi-hop schemes realise relay without *reputation-weighted* noise propagation, which is the distinguishing mechanism in our architecture. Recent learned-coordination schemes target adjacent problems: RALLY [52] is an LLM-driven role-adaptive yoked-navigation framework that outperforms multi-agent reinforcement-learning and LLM baselines on agentic UAV swarms, while CI-HRL [53] applies decentralised consensus-inference hierarchical reinforcement learning to multi-constrained UAV pursuit-evasion. Neither incorporates Byzantine resilience or trust-weighted state estimation. For load balancing, CLAC-CBBA [54] combines centrality-based cluster reconfiguration with load-aware self-regulation for heterogeneous swarms, but operates at the task-allocation layer without reputation weighting or Byzantine exclusion. For failover, Wu *et al.* [55] propose a QMIX/DQN-based multi-agent scheme that redeploys UAVs after mission-destruction events, without BFT, spoofing detection, or multi-hop relay. Finally, the concurrent Swarm Oracle work [56] demonstrates a blockchain-based BFT protocol on 12 physical robots at the theoretical $f=n/3$ limit, integrating a token-based reputation system that penalises both unreliable and inactive robots—but addresses collective perception for blockchain oracles rather than flight-targeted navigation under GNSS denial.

III. SYSTEM MODEL

Swarm composition. We consider a swarm $\mathcal{N} = \{v_1, \dots, v_n\}$ operating in a region without GNSS (or with partially spoofed GNSS). Each vehicle carries an IMU, a UWB ranging radio (accuracy ± 0.1 m, range 200–500 m), a communication transceiver, a computing unit with a secure key store, and a barometric altimeter. A subset $\mathcal{A}_{\text{cap}}$ ($|\mathcal{A}_{\text{cap}}|/n \approx 0.025\text{--}0.05$, i.e., 5–10 TRN-capable anchors per 200 UAVs) additionally carries a downward-facing camera and pre-stored satellite imagery, enabling terrain-relative navigation (TRN) with a learned feature extractor (e.g., SuperPoint [57]) and an incremental factor-graph optimiser (GTSAM/iSAM2 [58]). At any time each UAV is in one of three roles—*active anchor*, *follower*, or *standby anchor*—assigned by an anchor-score election (Section IV-E) and changeable in flight.

Threat model. The adversary can (a) capture up to f UAVs and replace their firmware while retaining valid cryptographic keys; (b) inject localised GNSS spoofing affecting a subset of the swarm [59]; (c) perform mass GNSS spoofing; (d) selectively jam links. The adversary cannot forge terrain descriptors for unobserved locations, cannot forge Ed25519 signatures without private keys, and cannot introduce Sybil identities [60] absent an out-of-band enrolment step. We distinguish *naïve* Byzantine peers (always falsify) from *strategic* peers (honest

TABLE I: Design-Space Coverage of Closest Prior Art. ✓ indicates published support; – indicates absent; (∼) indicates partial or mechanism-equivalent support. Columns: *BFT*, Byzantine fault tolerance; *Hier.*, anchor/follower hierarchy; *Repu.*, persistent reputation; *Fail.*, automatic failover; *Spoof.*, GNSS-spoofing detection; *Relay*, multi-hop position relay. SwarmRaft’s (∼) in the *Fail.* column reflects Raft leader re-election, which handles leader crash but does not redistribute followers or rebalance load upon leader loss; its (∼) in *BFT* reflects a Byzantine-resilient evaluation layer over a crash-fault Raft core. BFTL’s (∼) in *Fail.* reflects the formal route-completion guarantees under Byzantine leader loss proved in the BFTL paper, which constitute a form of leader-failure tolerance without an explicit follower-redistribution protocol. The *Repu.* column credits BFTL for blockchain-weighted route commitments (the BFTL mechanism), which differs from our reputation-weighted noise-covariance modulation of a continuous state estimator; the column entry records that a reputation-flavoured trust mechanism is present, not that the mechanisms are equivalent.

System	BFT	Hier.	Repu.	Fail.	Spoof.	Relay
SwarmRaft [12]	(∼)	–	–	(∼)	✓	–
BDGD [13]	✓	–	✓	–	✓	–
COUAVChain [26]	✓	✓	✓	–	–	–
BFTL [14]	✓	✓	✓	(∼)	–	✓
GTRI PNT Chain [1]	–	–	–	–	–	✓
Kimera-Multi [3]	–	–	–	–	–	✓
Omni-swarm [2]	–	–	–	–	–	✓
Swarm-SLAM [4]	–	–	–	–	–	✓
Moroncelli [38]	✓	–	✓	–	–	✓
Strobel [7]	✓	–	✓	–	–	–
Zhang <i>et al.</i> [39]	–	✓	–	–	–	(∼)
Yue <i>et al.</i> [40]	–	✓	–	–	–	✓
Mykytyn [45]	–	–	–	–	✓	–
This work	✓	✓	✓	✓	✓	✓

when observed, falsifying at critical moments); guarantees differ between these cases (Section V). Safety of quorum-based permanent exclusion requires $f \leq M - 1$, where M is the configurable quorum parameter (analysed in Section V).

Budgets and requirements. Per-UAV budget: <15 kbit/s bandwidth, <50 ms factor-graph optimisation on a ~ 100 -TOPS embedded accelerator. No external infrastructure (ground station, cloud, or blockchain validator). The swarm must maintain CEP <100 m for operational ISR, with continuous navigation under anchor destruction, Byzantine compromise, localised GNSS spoofing, and progressive attrition over 1–3 hour missions.

IV. PROPOSED APPROACH

We address the six pillars across seven subsections: Sections IV-B and IV-D establish BFT (Pillar 1); Section IV-C defines reputation (Pillar 3); Section IV-E covers the anchor/follower hierarchy (Pillar 2) and reputation-weighted multi-hop relay (Pillar 3); Section IV-F details automatic failover (Pillar 4); and Section IV-G addresses GNSS-spoofing detection (Pillar 5) and load balancing (Pillar 6). All hyperparameters introduced across these subsections are consolidated in Table II.

A. System Architecture Overview

Each UAV runs a common navigation module with four components: a *navigation core* performing visual-inertial

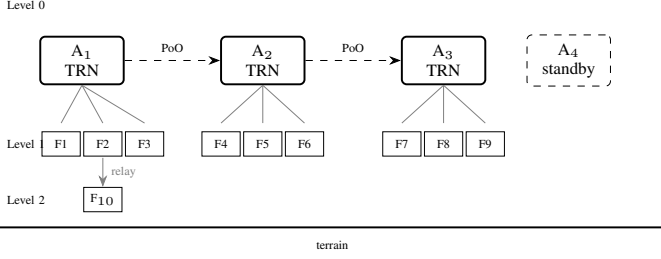


Fig. 1: Hierarchical architecture. Active anchors perform mutual Visual Proof of Observation (PoO) between one another and broadcast signed positions over UWB. Followers (Level 1) receive reputation-weighted position directly; deeper followers (Level 2+) receive relayed positions with per-hop noise growth. Standby anchors remain TRN-capable but are not currently elected.

odometry (VIO) at 50 Hz and TRN at 0.1 Hz (TRN only on anchors), fused in a local factor graph; an *observation recorder* that produces Signed Observations on each TRN fix; a *mesh protocol handler* managing signed-observation exchange, reputation tracking, and Byzantine rejection; and an *autopilot adapter* layer. The hierarchical topology overlays an anchor/follower structure on this substrate (Fig. 1): anchors perform PoO verification mutually, while followers receive reputation-weighted position via UWB relay with up to $K_{\max}$ hops.

B. Visual Proof of Observation

On each TRN fix at position $\mathbf{p}_i \in \mathbb{R}^3$ at time t_1 , UAV v_i extracts $N = 50$ top-scoring feature descriptors $\mathbf{f}_i \in \mathbb{R}^{N \times D}$ ($D=256$ for SuperPoint). $N = 50$ is a working default: retaining the top-50 SuperPoint keypoints by score [57] balances descriptor packet size (50 descriptors at 256 bytes each = 12.8 kB) against matching distinctiveness. Sensitivity to N is deferred to future work. The descriptor set forms the hash

$$H_i = \text{SHA-256}(\text{sort}_\downarrow(\mathbf{f}_i[0:N])). \quad (1)$$

It constructs the Signed Observation

$$\text{OBS}_i = (t_1, \mathbf{p}_i, H_i, c_i, \text{ID}_i, \sigma_i), \quad (2)$$

where c_i is the RANSAC inlier ratio and

$$\sigma_i = \text{Ed25519.sign}(sk_i, t_1 \|\mathbf{p}_i\| H_i \|c_i\| \text{ID}_i). \quad (3)$$

The base record (Mode B) totals approximately 180–200 bytes: t_1 (timestamp, 8 B) + $\mathbf{p}_i$ (position (lat, lon, alt) as three doubles, 24 B) + H_i (SHA-256 hash, 32 B) + c_i (RANSAC inlier ratio as a float, 4 B) + ID_i (UAV identifier, 8 B) + σ_i (Ed25519 signature, 64 B) = 140 B core payload, plus message framing (type byte, length, sequence number, CRC, padding) accounting for the remaining 40–60 B. A *Mode A* variant appends a compact feature set of $K = 20$ descriptors. The value $K = 20$ keeps Mode-A packet overhead within the UAV datalink budget assumed in Section II: descriptors are quantised from float32 to uint8 before transmission (256 dimensions, 1 B each), so the descriptor payload is $20 \times 256 = 5120$ B and the total Mode-A record is approximately 5.3 kB. Ed25519 [61] signature computation is well within sub-millisecond on modern processors at the cycle counts reported by [61] (87548 cycles for signing

on a Westmere CPU); typical ARM Cortex-class embedded targets fall in the same regime, with cycle-exact timings to be characterised on the target board in future work.

On receiving OBS_i , a neighbour v_j first verifies σ_i using pk_i . If the signature is invalid, reputation R_i is decreased by a signature penalty $\alpha_{\text{sig}} = 0.30$, a working default equal to $2 \times \alpha_{\text{neg}} = 0.30$ so a single signature mismatch has the same net reputation impact as two consecutive failed verifications. At a later time $t_2 > t_1$, when v_j flies within verification radius $r_{\text{verify}} = 500$ m of $\mathbf{p}_i$, it extracts local features $\mathbf{f}_j$ and computes a verification score. The value $r_{\text{verify}} = 500$ m matches the upper end of the UWB operational range (200–500 m at the datalink power levels assumed in Section II), ensuring that any follower within UWB communication range can request verification without requiring out-of-range RF links.

In *Mode A* (compact feature set), for each transmitted descriptor, the nearest neighbour in $\mathbf{f}_j$ is accepted if its L_2 distance is below $\tau_{\text{match}} = 0.7$ (Lowe’s ratio test threshold [62]: a threshold of 0.7 is the standard lower bound of the accepted range 0.7–0.8 for filtering ambiguous matches, used here with SuperPoint’s 256-dimensional L_2 -normalised descriptors [57]); the score V is the ratio of accepted matches to transmitted descriptors, and VERIFIED is declared when $V > T_{\text{verify}} = 0.3$. The threshold $T_{\text{verify}} = 0.3$ corresponds to 6/20 Mode-A descriptors (or 6/ N hash matches in Mode B) successfully passing the $\tau_{\text{match}} = 0.7$ Lowe-ratio test. Published cross-view SuperPoint matching studies report inlier ratios of 60–90% under clean line-of-sight conditions, falling to 20–40% under moderate viewpoint or illumination drift. Setting $T_{\text{verify}} = 0.3$ places the VERIFIED boundary at the lower end of the degraded-but-viable regime, so legitimate verifications continue to pass during moderate environmental noise while complete-failure cases (match ratios below $\sim 20\%$, characteristic of wrong-terrain or fabricated-hash submissions from a peer that has not physically observed the claimed region) are reliably rejected. The threshold interacts with τ_{match} and N, K : tightening any of the three has the same qualitative effect of raising the false-reject rate against environmentally-degraded honest verifications.

Sensitivity over $T_{\text{verify}} \in \{0.2, 0.3, 0.4\}$ is deferred to future work. Because $T_{\text{verify}} = 0.3$ sits at the lower end of the degraded-but-viable regime, honest peers operating under moderate environmental degradation can produce match scores near the exclusion boundary; combined with the asymmetric reputation rule (UNVERIFIED penalty $\alpha_{\text{neg}} = 0.15$), as few as $\lceil (R_{\text{init}} - T_{\text{reject}}) / (\alpha_{\text{neg}} R_k) \rceil = 2$ consecutive boundary-case UNVERIFIED verdicts under $R_k \approx 1$ would drive an honest peer’s reputation to T_{reject} . A formal false-acceptance / false-rejection (FAR/FRR) characterisation under realistic illumination, viewpoint, and seasonal terrain variation is left to a future calibration sweep; the corresponding empirical false-exclusion rate is reported in Limitations (Section XI). A *Mode B* variant (hash-only, zero-bandwidth verification) compares recomputed hashes directly; Mode B is suited to severely bandwidth-constrained tactical deployments in which flight conditions permit reproducible descriptor extraction, and requires additional calibration against the observation geometry (see Section XI).

The unforgeability rationale is physical: terrain feature descriptors are functions of the surface geometry, illumination, viewing angle, and the learned extractor. A captured UAV cannot fabricate descriptors for a location it has not physically observed without either possessing the imagery database and extractor (which still requires knowledge of the exact atmospheric and illumination conditions at t_1) or compromising honest peers before they verify—neither of which is feasible under our threat model. PoO is related to but distinct from the proof-of-location (PoL) literature: PoL systems such as VRChain [63] (blockchain-anchored visual homing) and BFT-PoLoc [64] (Byzantine-fortified trigonometric proof of location from Internet round-trip delays) verify a device’s *current* claimed position via either ledger consensus or signal-timing measurements. Visual PoO instead verifies a *past* observation through a neighbour’s later physical revisit, with evidence bound to terrain geometry via the learned extractor rather than to network timing or co-located witnesses. Because verification is retrospective (the verifier must fly within r_{verify} of the claimed location at $t_2 > t_1$), PoO is a slow reputation builder, not a real-time gate on the immediate observation: instantaneous protection against position lies is provided by the UWB consistency check of Section IV-G.

C. Asymmetric Reputation System

Each UAV maintains a reputation vector $\mathbf{R} = (R_1, \dots, R_n)$ with $R_j \in [0, 1]$ initialised to $R_{\text{init}} = 0.5$ (neutral prior). The initial reputation is set to the midpoint of the $[0, 1]$ range, representing maximum uncertainty about an unknown peer: neither trusted nor distrusted. This is the standard uninformative prior for reputation systems [65]. Updates on verification events are asymmetric:

$$\text{VERIFIED: } R_j \leftarrow \text{clamp}(R_j + \alpha_{\text{pos}} R_k, 0, 1), \quad (4)$$

$$\text{UNVERIFIED: } R_j \leftarrow \text{clamp}(R_j - \alpha_{\text{neg}} R_k, 0, 1), \quad (5)$$

$$\text{INVALID SIG: } R_j \leftarrow \text{clamp}(R_j - \alpha_{\text{sig}}, 0, 1), \quad (6)$$

where R_k is the reputation of the verifier. We require $\alpha_{\text{neg}} \geq 2\alpha_{\text{pos}}$, preferably $\alpha_{\text{neg}} \geq 3\alpha_{\text{pos}}$; the default values are $\alpha_{\text{pos}} = 0.05$, $\alpha_{\text{neg}} = 0.15$. The 3:1 ratio ($\alpha_{\text{neg}}/\alpha_{\text{pos}}$) is set in Section V-B; the absolute level is anchored by the convergence bound of Proposition 3: to achieve $T_{\text{ind}} \approx 170$ s with $R_{\text{init}} = 0.5$, $T_{\text{reject}} = 0.2$, and $\nu_{\text{verify}}^{-1} \approx 45$ s, the approximation requires $\alpha_{\text{neg}} \cdot R_{\text{honest}} \cdot \nu_{\text{verify}} \geq (R_{\text{init}} - T_{\text{reject}})/T_{\text{ind}} \approx 0.30/170 \approx 0.0018$ per second, or $\alpha_{\text{neg}} \geq 0.0018 \times 45 \approx 0.08$ per verification event; $\alpha_{\text{neg}} = 0.15$ satisfies this with a $\approx 2\times$ margin. The value $\alpha_{\text{pos}} = 0.05$ follows from the 3:1 ratio. Under this asymmetry, three failed verifications outweigh nine successful ones ($3 \cdot 0.15 = 9 \cdot 0.05 = 0.45$), guaranteeing that distrust accumulates faster than trust while retaining a margin of 2–8 consecutive environment-induced failures before an honest peer is falsely excluded (2 pure-burst events from $R_{\text{init}} = 0.5$ at $R_k = 1.0$; up to 8 interleaved VERIFIED+UNVERIFIED rounds from $R_j = 1.0$; derivation follows). Let R_j be the *target* peer’s current reputation and R_k the *verifier*’s reputation weight. Each UNVERIFIED event decrements R_j by $\alpha_{\text{neg}} \times R_k$; each VERIFIED event increments R_j by α_{pos} . For pure-UNVERIFIED bursts (worst case), events needed to

drive R_j to $T_{\text{reject}} = 0.2$ are $\lceil (R_j - T_{\text{reject}})/(\alpha_{\text{neg}} \times R_k) \rceil$. At $R_k = 1.0$ and $R_j = R_{\text{init}} = 0.5$: $\lceil 0.3/0.15 \rceil = 2$; at $R_j = 1.0$: $\lceil 0.8/0.15 \rceil = 6$. For interleaved sequences (one VERIFIED plus one UNVERIFIED per round), net loss per round is $\alpha_{\text{neg}} \times R_k - \alpha_{\text{pos}}$; at $R_k = 1.0$, $0.15 - 0.05 = 0.10$, giving 3 rounds from $R_j = 0.5$ to 8 rounds from $R_j = 1.0$. Overall margin: 2 (pure burst from R_{init}) to 8 (interleaved from $R_j = 1.0$) consecutive failures.

To handle dynamic environments, we apply a temporal decay

$$V_{\text{weighted}} = V \cdot \exp(-\lambda \Delta t), \quad (7)$$

with $\lambda = 10^{-3} \text{ s}^{-1}$ (half-life ≈ 11.5 min). The decay constant is chosen so that an observation becomes effectively stale ($< 5\%$ residual weight) after ≈ 3 half-lives = 34.5 min, approximately one-tenth of the 2-hour mission (the one-tenth fraction is an operationally discretionary design goal: it ensures that any single mission phase does not permanently dominate the reputation estimate, while still allowing short-horizon events to decay within one mission); a temporary gap of one revisit cycle ($\nu_{\text{verify}}^{-1} \approx 45$ s) causes only $\approx 4.4\%$ decay per missed event ($e^{-10^{-3} \times 45} \approx 0.956$). These are working defaults pending a future sensitivity analysis. Observations older than $T_{\text{max_verify}} = 3600$ s (1 hour) are considered stale. This cutoff is set to half the 2-hour planned mission length, ensuring that observations from the first half of a mission are discarded before the second half begins. For shorter or longer missions the cutoff should be scaled proportionally.

D. Reputation-Weighted Distributed Factor Graph

Each UAV maintains a local factor graph $\mathcal{G}_i$ whose maximum-a-posteriori estimate is

$$\mathbf{X}_i^* = \arg \min_{\mathbf{X}} \sum_k \|h_k(\mathbf{X}) - \mathbf{z}_k\|_{\Sigma_k}^2. \quad (8)$$

Three classes of factors are instantiated:

- *Local factors* (IMU preintegration [66], VIO, TRN-absolute on anchors, barometric altitude), standard in the art.
- *Ranging factors* for each UWB measurement d_{ij} with $h_{\text{range}}(\mathbf{X}_i, \mathbf{X}_j) = \|\mathbf{p}_i - \mathbf{p}_j\|_2$, $\sigma_{\text{range}} = 0.1$ m.
- *Reputation-weighted priors* from the Distilled State of neighbour j , where the *Distilled State* of UAV j is the per-UAV broadcast tuple $\text{DS}_j = (t, \mathbf{p}_j, \mathbf{v}_j, \text{cov}(\mathbf{p}_j), R_j, \text{ID}_j, \sigma_j^{\text{Ed25519}})$ comprising timestamp, position, velocity, position-covariance summary, current reputation, identifier, and Ed25519 signature; the byte-level payload composition and bandwidth accounting are given in Section V-D. The Distilled State is the central inter-UAV communication primitive of the architecture and is consumed by the reputation-prior factor here, by the failover watchdog (Section IV-F, phase F1), and by the load-balancing broadcast (Section IV-G). The reputation-prior factor uses the position $\mathbf{p}_j$ and covariance $\text{cov}(\mathbf{p}_j)$ of DS_j , scaled by the recipient’s locally-maintained reputation $R_j^{(\text{local})}$ of sender j (not the self-reported R_j field of the

broadcast tuple; the broadcast field is advisory/diagnostic only). This preserves the no-gossip assumption of Proposition 1 assumption (iii): the reputation that determines weighting is the verifier’s own, never the target’s self-reported value. The same rule applies to every reputation-weighted decision in the architecture: the reputation term in the anchor score (Eq. 10) and in the failover-offer score (Eq. 13) is the evaluator’s *local* reputation of the candidate, not its self-reported value. In particular, the reputation an anchor includes in a REASSIGN_OFFER (Section IV-F, phase F3) is advisory only; the orphaned follower scores each offer with its *own* local reputation of the offering anchor, so a Byzantine anchor cannot attract followers by claiming inflated reputation. The broadcast reputation field of DS_j is therefore diagnostic/telemetry only and is never consumed by a weighting or selection rule. Noise

$$\sigma_j = \sigma_{\text{base}} / (R_j + \varepsilon), \quad (9)$$

$\sigma_{\text{base}} = 50$ m, $\varepsilon = 0.01$ (dimensionless numerical regularisation; chosen two orders of magnitude below $\sigma_{\text{base}} = 50$ m so $\varepsilon/\sigma_{\text{base}} = 0.02\%$, ensuring negligible influence on the computed covariance). Covariance $\Sigma_j = \text{diag}(\sigma_j^2, \sigma_j^2, (\sigma_j/2)^2)$ for (lat, lon, alt); the halved altitude term reflects a standard anisotropy assumption that UAV barometric altimeters provide tighter altitude bounds than UWB-derived lateral uncertainty. A dedicated altitude/lateral-ratio characterisation against measured flight data is left to future work. Sensitivity to σ_{base} : the analytical CEP bound in the multi-hop error model (Section V-C) scales linearly with σ_{base} ; doubling to 100 m would require approximately $2\times$ as many anchor observations to converge to the 100 m CEP pass criterion. A full sensitivity sweep over $\sigma_{\text{base}} \in \{25, 50, 75, 100\}$ m is planned in future work.

A UAV with $R = 1.0$ contributes $\sigma \approx 49.5$ m, versus $\sigma \approx 238$ m at $R = 0.2$ —approximately a $5\times$ weight ratio in precision and $\sim 23\times$ in variance. Optimisation uses iSAM2 with the Bayes-tree representation; the <50 ms per-update latency on an embedded accelerator is a design target consistent with published GTSAM/iSAM2 benchmarks [58]. The <50 ms budget covers the iSAM2 optimisation step only; the full PoO pipeline (camera capture ~ 10 ms, SuperPoint inference ~ 20 – 40 ms on a 50–100 TOPS-class accelerator at 480×640 resolution per published embedded benchmarks [57], top- N sort <1 ms, SHA-256 hash <1 ms, and Ed25519 signing <1 ms at the 87 548-cycle Westmere figure from [61] on an ARM Cortex-class core) is constrained by the 0.1 Hz TRN fix rate ($=10$ s budget) rather than by the 50 ms factor-graph cycle, so SuperPoint inference timing is not on the critical sub-50 ms path. Per-board characterisation of the PoO pipeline on the specific target accelerator is a calibration item left to future work; the 50–100 TOPS class is named here as a representative envelope rather than a specific SKU.

A UWB consistency check provides a secondary, non-visual channel: if UAV j reports position $\mathbf{p}_j^{\text{rep}}$ but UWB ranging d_{ij} yields $\|\mathbf{p}_i - \mathbf{p}_j^{\text{rep}}\| - d_{ij} > \delta_{\text{consist}} = 50$ m, reputation decreases by $\alpha_{\text{range}} = 0.10$. The threshold $\delta_{\text{consist}} = 50$ m

matches $\sigma_{\text{base}} = 50$ m (the declared prior noise), so a consistency check flags deviations beyond one prior standard deviation. The design value $T_{\text{spool}} = 50$ m (Section IV-G) is calibrated against the same noise floor; the Phase 5 production sweep uses a tighter operational threshold (Section VI-F). The range-failure penalty $\alpha_{\text{range}} = 0.10$ is set as the midpoint $(\alpha_{\text{pos}} + \alpha_{\text{neg}})/2$, treating a UWB consistency failure as more severe than a single missed verification (α_{pos}) but less severe than a full failed verification (α_{neg}), since range failures may arise from temporary link degradation rather than deliberate Byzantine behaviour.

E. Dynamic Anchor Selection and Position Relay

TRN-capable UAVs compete for anchor status through the score

$$A_i = w_{\text{rep}} R_i + w_{\text{conf}} C_i + w_{\text{sat}} S_i + w_{\text{fol}} F_i, \quad (10)$$

with default weights (0.4, 0.3, 0.2, 0.1). Here C_i is the TRN confidence (RANSAC inlier ratio), S_i is the imagery coverage quality for the terrain below v_i , and F_i is an inverse load factor. The weight ordering $w_{\text{rep}} > w_{\text{conf}} > w_{\text{sat}} > w_{\text{fol}}$ reflects a safety-first hierarchy: trustworthiness must dominate performance, which in turn dominates location-dependent factors (imagery quality varies with where the UAV currently is, not what it is), with load balancing as the lowest-priority factor since over-loading is a soft degradation rather than a safety violation. Placing $w_{\text{rep}} = 0.4$ as the dominant term ensures that a compromised peer with degraded reputation $R_i \rightarrow 0$ contributes at most 0.6 to its anchor score even with maximal C_i, S_i, F_i , leaving it below honest candidates whose $R_i \geq T_{\text{quorum}} = 0.6$ adds at least 0.24 from the reputation term alone—an inherent score gap that makes Byzantine-anchor promotion infeasible without first restoring reputation through honest verifications. Given this ordering and the unit-sum constraint, the equal-spacing choice (0.4, 0.3, 0.2, 0.1) is adopted as a working default; the spacing $\Delta w = 0.1$ exceeds the per-event reputation step $\alpha_{\text{pos}} = 0.05$, so single-verification reputation jitter cannot swap adjacent priority terms in the score ordering. Sensitivity to the weight vector is part of a planned ablation sweep over $w_{\text{rep}} \in [0.3, 0.5]$, $w_{\text{conf}} \in [0.2, 0.35]$, $w_{\text{sat}} \in [0.15, 0.30]$, $w_{\text{fol}} \in [0.05, 0.20]$ subject to the unit-sum constraint.

Every $T_{\text{election}} = 60$ s, the M vehicles with the highest A_i (Eq. 10) are confirmed as active anchors; a standby replaces an active anchor when its score exceeds the incumbent’s by a hysteresis $H = 0.1$. The election period $T_{\text{election}} = 60$ s exceeds the verification revisit cycle ($\nu_{\text{verify}}^{-1} \approx 45$ s) by 33%, allowing at least one full verification round to complete between elections and preventing frequent re-elections from disrupting ongoing verifications while remaining responsive to anchor degradation. The hysteresis band $H = 0.1$ corresponds to two consecutive $\alpha_{\text{pos}} = 0.05$ upward reputation steps, requiring the score gap to represent at least two successful-verification equivalents before triggering a handover. Handover proceeds in three steps (*PROMOTION*, *DEMOTION*, linear-blend) over $T_{\text{handover}} = 10$ s, matching the failover total budget of Section IV-F (F1–F5 phases sum to ≈ 8 s nominal, 10 s

TABLE II: Consolidated system hyperparameters. Derivation column indicates whether the value is literature-derived (L), operationally discretionary (O), or a working default pending sensitivity analysis (W).

Parameter	Value	Symbol	Derivation
Feature descriptors per PoO	50	N	W
Mode-A compact set size	20	K	W
Signature penalty	0.30	α_{sig}	W
Verification radius	500 m	r_{verify}	W
L2 match threshold	0.7	τ_{match}	L (SuperPoint)
VERIFIED score threshold	0.3	T_{verify}	W
Initial reputation	0.5	R_{init}	W
Positive reputation step	0.05	α_{pos}	W
Negative reputation step	0.15	α_{neg}	W (3:1 ratio)
Temporal decay constant	10^{-3} s^{-1}	λ	W
Staleness cutoff	3600 s	$T_{\text{max_verify}}$	O
Base prior noise	50 m	σ_{base}	W
Noise floor	0.01	ε	W
UWB consistency threshold	50 m	δ_{consist}	W
Range-failure penalty	0.10	α_{range}	W
Anchor-score weights	(0.4,0.3,0.2,0.1)	$w_{1..4}$	W
Election period	60 s	T_{election}	O
Standby-replace hysteresis	0.10	H	W
Handover blend duration	10 s	T_{handover}	O
Sybil warmup count	6	T_{warmup}	W
Per-hop noise growth	0.15	β	W
Max relay hops (ISR)	3	K_{max}	O
Max relay hops (patrol)	5	K_{max}	O
Reject threshold	0.20	T_{reject}	W
Quorum reputation floor	0.60	T_{quorum}	W
Load-balance period	30 s	T_{balance}	O
Load-balance low threshold	30%	P_{low}	O
Load-balance high threshold	150%	P_{high}	O
Spoof position threshold	50 m	T_{spooof}	W
Consecutive spoof count	3	$N_{\text{consecutive}}$	W
Mass-spoofing trigger	30%	P_{mass}	O

with 5% packet loss); all anchor-change operations—whether triggered by destruction or by score-based replacement—share the same architectural timescale.

The lower bound on T_{handover} is set by the cumulative protocol latency of the three-step PROMOTION $\rightarrow$ DEMOTION $\rightarrow$ linear-blend sequence (UWB neighbour discovery ~ 1 s, role-change acknowledgement round-trip ~ 1 – 2 s, follower-side covariance update propagation through the iSAM2 factor graph ~ 1 s), so T_{handover} must exceed ~ 4 s to ensure all followers complete their re-association within the blend window. The upper bound is set by the verification revisit cycle: T_{handover} must remain well below $\nu_{\text{verify}}^{-1} \approx 45$ s so that handovers are decisive on a timescale shorter than reputation reassessment. During the blend, both incumbent and standby publish Distilled State updates with time-weighted weights summing to unity; a longer blend would increase UWB neighbour bandwidth consumption and follower-side factor-graph merge events, while a shorter blend risks discontinuities between the demoted-incumbent’s last publication and the promoted-standby’s first authoritative state. Sybil protection requires $T_{\text{warmup}} = 6$ successful verifications before anchor eligibility; at $\nu_{\text{verify}}^{-1} \approx 45$ s per cycle, this is ≈ 4.5 min of observed honest behaviour, exceeding four full election periods, so a freshly-joining Byzantine peer cannot accumulate quorum influence within a single election cycle.

Followers compute their position as the reputation-weighted

average

$$\mathbf{p}_{\text{fol}} = \frac{\sum_j w_j \mathbf{p}_j}{\sum_j w_j}, \quad w_j = \frac{R_j}{\sigma_{\text{hop},j}^2}, \quad (11)$$

where for a k -hop relay chain,

$$\sigma_k = \sigma_{\text{anchor}} (1 + \beta)^k, \quad (12)$$

with $\beta = 0.15$ per hop (per-hop noise growth factor, a conservative working default; see Section III discussion) and σ_{anchor} given by (9). The multiplicative form $\sigma_k = \sigma_{\text{anchor}} (1 + \beta)^k$ models two physical effects that compound at each relay hop in a UWB-trilaterated position chain: (i) additive UWB ranging-measurement variance σ_{range}^2 from the new measurement to the previous hop, and (ii) propagated upstream position uncertainty from the previous hop’s estimate, weighted by a Geometric Dilution of Precision (GDOP) factor that depends on the local anchor configuration. Under perfectly independent ranging and ideal anchor geometry, error propagation is sublinear (*additive variance model*): $\sigma_k = \sqrt{\sigma_{\text{anchor}}^2 + k\sigma_{\text{range}}^2}$,

growing as $\sqrt{k}$ for large k —this is the theoretical lower bound for independent Gaussian ranging noise. The multiplicative form is not a physical prediction but a *conservative engineering bound*: it dominates the additive model for all $k \geq 1$ when $\beta > 0$, as $(1 + \beta)^k > \sqrt{1 + k(\sigma_{\text{range}}/\sigma_{\text{anchor}})^2}$ for typical UWB parameters. Under correlated upstream uncertainty or poor geometry (anchors nearly collinear with the follower), GDOP amplifies propagated variance beyond the sublinear baseline, with the worst case approaching exponential $(1 + \beta)^k$ growth. The multiplicative form is therefore the *conservative engineering bound* for the planned non-collinear but topologically constrained swarm. The value $\beta = 0.15$ is adopted as a conservative working default for the per-hop standard-deviation growth factor; it represents a 15% std-dev increase per hop relative to the prior hop, chosen as a round figure within the typical range expected for UWB cross-trilateration under benign (line-of-sight) propagation and pending empirical refit against a measured UWB NLOS dataset in future work. From this β , the $K_{\text{max}} = 3$ ISR ceiling follows as a derived quantity (at the reference operating point $\sigma_{\text{anchor}} \approx 58$ m, $R_{\text{anchor}} = 0.85$, $\sigma_3 \approx 88$ m, giving single-anchor $\text{CEP}_{50} = 1.177\sigma_3 \approx 104$ m, slightly exceeding the 100-m operational ISR-grade target by $\approx 4\%$; under multi-anchor reputation-weighted fusion ($m=2$ – 3) the bound improves to $\text{CEP}_{50} \in [60, 74]$ m, comfortably within the target); $K_{\text{max}} = 3$ is not used to motivate the choice of β . A sensitivity sweep over $\beta \in \{0.05, 0.10, 0.15, 0.20\}$ is planned in future work and will shift the derived K_{max} accordingly. The functional form and value of β are subject to empirical refit in future work: if measured σ_k versus k fits a sublinear $\sqrt{k}$ form or a different multiplicative rate β' better than the conservative $(1 + 0.15)^k$ assumption, the K_{max} relay-depth ceiling derived above (Section V-C) relaxes accordingly without affecting the analytical structure. The base noise σ_{base} is a configurable system parameter; we adopt $\sigma_{\text{base}} = 50$ m for consistency across the experiments reported below, with other values applicable to swarm configurations that assume different baseline anchor accuracy. For a well-performing anchor with $R_{\text{anchor}} = 0.85$, $\sigma_{\text{anchor}} = 50/0.86 \approx 58.1$ m; propagation yields $\sigma_1 \approx 66.8$, $\sigma_2 \approx 76.9$, $\sigma_3 \approx 88.4$, and

$\sigma_5 \approx 116.9$ m. The 5-hop value exceeds the 100-m target both as σ_5 and as the corresponding $\text{CEP}_{50} \approx 138$ m; for operational ISR we therefore recommend $K_{\max} = 3$ under the default multi-anchor fusion regime, reserving $K_{\max} = 5$ for lower-precision patrol missions. The patrol value $K_{\max} = 5$ is reported as an upper operational envelope, not a precision recommendation: at $R_{\text{anchor}} = 0.85$ the per-hop derivation yields $\sigma_5 \approx 117$ m ($\text{CEP}_{50} \approx 138$ m), exceeding the 100 m operational ISR target. The intermediate $K_{\max} = 4$ is not separately reported because it is not operationally distinct from $K_{\max} = 5$ for precision purposes; $K_{\max} = 3$ under multi-anchor fusion is the maximum strictly ISR-compliant depth (single-anchor at $k = 3$ exceeds the CEP target by $\approx 4\%$ and is flagged for deployment-time review). The patrol use case accepts precision on the order of ~ 150 m CEP or coarser in exchange for a $1.67\times$ greater effective swarm reach (relay radius 2.5 km at $K_{\max} = 5$ versus 1.5 km at $K_{\max} = 3$, given the 500-m UWB hop range). For anchors operating below the reference $R_{\text{anchor}} = 0.85$, σ_k scales as $1/(R_{\text{anchor}} + \varepsilon)$ per (9), so both bounds tighten proportionally; the precision tolerance is a per-mission operator decision and the relay-depth ceiling is set deployment-time as a function of the mission's precision class. If R_{anchor} drops from 0.85 to 0.30, σ_{anchor} rises to ~ 161 m and all downstream followers automatically receive degraded weights through (11) without any explicit control signal: *reputation flows down the hierarchy through noise covariance*.

Empirical refit. The empirical campaign of Section VI-C reports a 250-trial measurement of CEP versus relay depth on a triangulated UWB graph under paper-spec sensor noise ($\sigma_{\text{UWB}} = 0.1$ m, $\sigma_{\text{TRN}} = 35$ m, TRN period 10 s) with realistic IMU-style initial-noise on both anchors ($\sigma_{\text{init}} = 30$ m) and followers ($\sigma_{\text{init}} = 50$ m). Two functional forms for σ_k are statistically indistinguishable at this sample size ($\Delta\text{AIC} < 1$ between fits): a linear form $\text{CEP}(k) \approx 19.6 + 1.89k$ m and the GDOP-family exponential of (12) with $\sigma_0 \approx 19.4$ m and $\delta_{\text{empirical}} = 0.088$. The symbol $\delta_{\text{empirical}}$ denotes the same per-hop multiplicative growth factor as the design-time β in Eq. (12); the distinct notation marks that one is a measured fit and the other a conservative design assumption. The GDOP form is adopted for continuity with the paper's original formulation. The conservative $\beta = 0.15$ assumed above is confirmed as a true upper bound: the measured $\delta_{\text{empirical}} = 0.088$ is comfortably inside $\beta = 0.15$ ($1.088 < 1.15$ per-hop multiplier), and the empirical relay envelope at $k=3$ ($\text{CEP} = 23.3$ m) lies more than $4\times$ inside the 100-m operational ISR target. Sparse-chain extension (single-relay-per-hop topology) is identified as a follow-on study.

F. Automatic Failover Protocol

Anchor loss (destruction, jamming, or reputation falling below T_{reject}) triggers a five-phase protocol executing within 10 s:

- **F1 Detection** (0–5 s): a watchdog declares anchor unavailability after $T_{\text{timeout}} = 5$ s without a DistilledState message. During the timeout, orphaned followers propagate by IMU dead reckoning, accumulating < 5 m drift at 30 m/s.

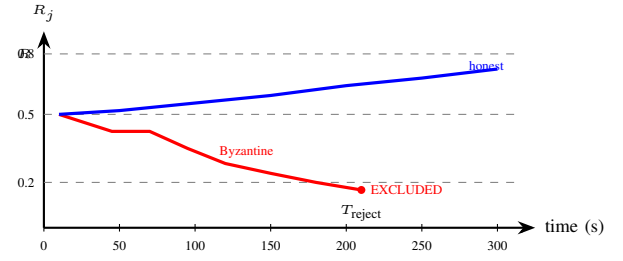


Fig. 2: Schematic of the analytically predicted reputation trajectory under a naïve Byzantine attack (Proposition 3; the schematic is an analytical prediction; empirical validation of the reputation-driven exclusion mechanism is reported in Section VI-E with 30/30 trials and zero false honest exclusions). The compromised UAV's reputation decays after each failed Proof-of-Observation verification while honest peers continue to accumulate trust. Permanent exclusion is predicted within the parametric estimate $T_{\text{exclusion}} \approx 215$ s under the conservative parameters of Proposition 3; the schematic marks $T \approx 210$ s, at which three independent honest verifiers have each driven R_j below $T_{\text{reject}} = 0.2$. The per-step drop $R_4 : 0.5 \rightarrow 0.425$ illustrated at $T = 45$ s corresponds to an illustrative verifier weight $R_k = 0.5$ (so that $\alpha_{\text{neg}} \cdot R_k = 0.075$); the operational deployment uses R_k drawn from the honest reputation distribution, and the Proposition 3 estimate uses the conservative substitution $R_k \rightarrow \bar{R}_{\text{honest}} = 0.53$ for the analytical derivation.

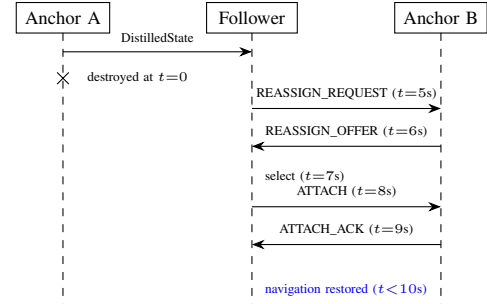


Fig. 3: Failover sequence upon anchor destruction. Dead-reckoning drift during the 5-second timeout is < 5 m at 30 m/s cruise speed.

- **F2 REASSIGN_REQUEST** (5–6 s): each orphaned follower broadcasts its device ID, last known position, velocity, battery state, and sensor capabilities.
- **F3 REASSIGN_OFFER** (6–7 s): surviving anchors respond with their ID, position, an advisory reputation field, and available capacity; the orphaned follower scores each offer using its *own* local reputation of the offering anchor (Section IV-D), so a Byzantine anchor cannot attract followers by inflating the advisory field.
- **F4 Selection** (7–8 s): the follower computes

$$S_{\text{reassign}} = \alpha_{\text{prox}}/d + \alpha_{\text{rep}}R + \alpha_{\text{cap}}(1 - L), \quad (13)$$

with defaults (0.4, 0.35, 0.25).

- **F5 Attachment** (8–10 s): ATTACH/ATTACH_ACK exchanges position and covariance.

Byzantine rejection proper removes all factors of UAV j from the local graph when $R_j < T_{\text{reject}} = 0.2$; permanent exclusion additionally requires M independent UAVs, each with $R > T_{\text{quorum}} = 0.6$, to confirm the exclusion. From Proposition 3, convergence is predicted at $T_{\text{ind}} = (R_{\text{init}} - T_{\text{reject}})/(\alpha_{\text{neg}} \bar{R}_{\text{honest}} \nu_{\text{verify}}) \approx 170$ s; with $R_{\text{init}} = 0.5$ the gap $R_{\text{init}} - T_{\text{reject}} = 0.3$ drives the numerator. A smaller T_{reject} would increase convergence time, while a larger T_{reject} would increase false exclusions of honest peers; $T_{\text{reject}} = 0.2$ keeps

the honest-peer safety margin above 2 environment-induced failures at $R_k = 1.0$ (and above 4 at $R_k = 0.5$; derivation in Section IV-C). The quorum reputation floor $T_{\text{quorum}} = 0.6$ requires a verifier to have earned at least two successful verifications above the initial state ($T_{\text{quorum}} = R_{\text{init}} + 2\alpha_{\text{pos}} = 0.5 + 0.10 = 0.6$), so a freshly-joined peer cannot self-confirm into the quorum; the minimum defensible value is $T_{\text{quorum}} > R_{\text{init}} = 0.5$, and $T_{\text{quorum}} = 0.6$ provides a minimal earned-trust barrier. Coincidentally $T_{\text{quorum}} = 3 \times T_{\text{reject}}$, but the binding constraint is the margin above R_{init} , not the factor of 3; the factor of 3 is a working default providing a clear separation between verifier quality and exclusion threshold. The default $M = 3$ tolerates $f \leq M - 1 = 2$ colluding Byzantine peers irrespective of n ; safety for larger f requires M to scale with f , as analysed in Section V. Fig. 3 summarises the timing.

G. Load Balancing and GNSS Spoofing Detection

Every $T_{\text{balance}} = 30$ s, anchors broadcast a STATUS field (appended to the standard Distilled State message of Section IV-D) including their follower count. The load-balancing period is set to one-half of the election period ($T_{\text{election}} = 60$ s), allowing two load-balance checks per election cycle; at $\nu_{\text{verify}}^{-1} \approx 45$ s, a 30 s check interval falls between consecutive verifications. The target is $T_{\text{target}} = N_{\text{total}}/N_{\text{anchors}}$. If an anchor's count is below $P_{\text{low}}T_{\text{target}}$ (30%) or above $P_{\text{high}}T_{\text{target}}$ (150%), a redistribution protocol (TRANSFER_OFFER $\rightarrow$ TRANSFER_ACCEPT $\rightarrow$ REASSIGN_COMMAND $\rightarrow$ smooth linear-blend handover over $T_{\text{handover}} = 10$ s) transfers followers from overloaded to underloaded anchors. The asymmetric band reflects the asymmetric cost of under- vs over-loading: an overloaded anchor ($> 150\%$ of target) risks queue overflow and message drops, while an underloaded anchor ($< 30\%$ of target) wastes TRN capacity without causing message loss. Overload headroom: $P_{\text{high}} - 1.0 = 0.50$; underload floor: $1.0 - P_{\text{low}} = 0.70$. These are working defaults consistent with general load-balancing practice [67].

Admission control.: Bring-up of the load-balancing pipeline in Phase 6 (Section VI-F) exposed a thundering-herd anti-pattern: simultaneously-displaced followers receiving offers from M survivors after an attrition event compute the same selection score and converge on the same anchor, opening a ~ 30 s over-capacity window before the protocol self-corrects on the next event. Two cooperating mitigations are added to the protocol and are present in all empirical numbers reported herein: (i) *anchor-side admission control*—an anchor refuses an ATTACH_REQUEST when $n_{\text{attached}} \geq \text{target_capacity}$, deferring the would-be attacher to its next-best ranked offer; and (ii) *follower ATTACH-timeout retry*—if the expected ATTACH_ACK does not arrive within $t_{\text{att}} = 2$ s, the follower tries the next anchor in its ranked-offer list. With both patches the over-capacity window collapses to zero (Section VI-F, Table VI), at the cost of one displaced follower per 15 trials that exhausts its fallback list before re-attaching—an explicit Pareto trade-off between capacity-invariant enforcement and mission-attach success, reported as such in the empirical campaign.

GNSS-spoofing detection operates at 1 Hz on each UAV with an active GNSS receiver (relevant under partial denial or during transition to full denial). The vehicle compares its GNSS-reported position with a UWB-trilaterated position from at least three neighbours with $R > T_{\text{min}} = T_{\text{reject}} = 0.20$. If $\|\mathbf{p}_{\text{GNSS}} - \mathbf{p}_{\text{UWB}}\| > T_{\text{spooft}} = 50$ m for $N_{\text{consecutive}} = 3$ successive measurements, GNSS is marked unreliable and the vehicle falls back to reputation-weighted UWB position. The design value $T_{\text{spooft}} = 50$ m matches $\sigma_{\text{base}} = 50$ m, flagging deviations beyond one prior standard deviation; at 500 m UWB range with typical 10–30 cm ranging error, this is ~ 170 – $500\sigma_{\text{UWB}}$, far above any legitimate uncertainty. $N_{\text{consecutive}} = 3$ reduces per-sample false-alarm probability p to approximately p^3 while staying below the $T_{\text{warmup}} = 6$ threshold. The empirical Phase 5 sweep (Section VI-F) uses a tighter operational threshold $T_{\text{spooft}} = 20$ m to characterise detector latency near the noise floor; the 50 m design value remains the spec for deployments seeking minimum false-positive risk.

When the proportion of spoofed UAVs exceeds $P_{\text{mass}} = 30\%$, the entire swarm transitions to a *full GNSS-denial mode*: GNSS is disabled globally and the swarm relies exclusively on TRN anchors with UWB relay. This threshold sits just below the classical Byzantine bound $f < 1/3$ and well above the per-Proposition-1 maximum absolute Byzantine count $f \leq 2$ at default $M = 3$ ($\approx 1\%$ at $n = 200$); a coordinated attack on $> 30\%$ of the swarm falls outside the individual-exclusion model and triggers the global response. $P_{\text{mass}} = 30\%$ is operator-configurable.

By design, GNSS-spoof events do not affect navigation reputation R_i : a UAV spoofed by an external emitter is not punished for an attack it did not launch. A captured peer that spoofs only its GNSS receiver while continuing to fabricate position claims is still excluded through two orthogonal channels: the Proof-of-Observation verification of Section IV-B and the UWB consistency check above, both of which feed the reputation-update rule.

V. THEORETICAL ANALYSIS

A. Quorum Safety

Proposition 1 (Quorum safety against naïve Byzantine peers): Let M be the permanent-exclusion quorum size. Assume (i) naïve Byzantine peers fail every Proof-of-Observation verification (they claim positions they have not physically observed); (ii) honest peers correctly issue VERIFIED or UNVERIFIED verdicts; (iii) each UAV maintains its reputation state R_j locally from its own verifications, without gossip-based propagation of reputation values between peers; and (iv) the per-peer rate of environment-induced UNVERIFIED verdicts on honest targets, ρ_{env} (events per second), satisfies $\rho_{\text{env}} < \alpha_{\text{pos}} \nu_{\text{verify}} / \alpha_{\text{neg}}$, so that honest peers accumulate VERIFIED events faster than environmental penalties drive R toward T_{reject} . Condition (iv) is a *rate condition on the environmental failure regime*; the Remark below quantifies what happens when it is violated (e.g., persistent cloud cover or night-time operation). The proposition provides no safety guarantee against environment-induced false exclusion when (iv) is violated, only against deliberate Byzantine attack; and

(v) $R_{\text{init}} < T_{\text{quorum}}$, so that a freshly-joined Byzantine peer cannot be self-confirmed into the quorum by virtue of its initial reputation alone (the default values $R_{\text{init}} = 0.5 < T_{\text{quorum}} = 0.6$ satisfy this). If $f \leq M - 1$, no set of colluding Byzantine peers can trigger false permanent exclusion of an honest UAV.

Sketch: Permanent exclusion requires M independent UAVs, each with $R > T_{\text{quorum}}$, to declare the target $R_j < T_{\text{reject}}$. We show no false quorum can form against an honest target. By assumption (i), every verification by an honest verifier v_k against a Byzantine target v_j yields UNVERIFIED; by the reputation update rule of Eq. (5), each such event monotonically decreases R_j at v_k (which is the only R value the rule modifies, by assumption (iii) — no gossip-based propagation). Hence R_j at any honest verifier can never strictly exceed its initial value R_{init} . By assumption (v), $R_{\text{init}} < T_{\text{quorum}}$. Therefore no honest verifier ever sees a Byzantine peer's reputation cross T_{quorum} , and no Byzantine peer can vote in the quorum. Any false quorum must therefore consist entirely of honest verifiers; for an honest target this requires at least one of them to issue a false UNVERIFIED, contradicting assumption (ii). Assumption (iv) further bounds the environment-induced UNVERIFIED rate so that an honest target's reputation does not drift below T_{reject} from environmental causes alone, eliminating the alternative failure mode. ■

Remark 1 (Boundary case: post-accumulation compromise): Proposition 1 as stated covers peers that enter Byzantine behaviour from $R_j \leq R_{\text{init}} < T_{\text{quorum}}$. The operationally important boundary case — a previously-honest peer that accumulated $R_j > T_{\text{quorum}}$ through legitimate verifications and is subsequently captured — is not covered by the safety guarantee at the instant of compromise. The system, however, drains the compromised peer's quorum eligibility through three independent mechanisms: (i) asymmetric reputation decay (Eq. 5), which subtracts α_{neg} per failed PoO verification at the post-compromise UNVERIFIED rate; (ii) the UWB consistency check of Section IV-D, which penalises range/position inconsistency at rate α_{range} independently of PoO; and (iii) the temporal decay of Eq. (7) with $\lambda = 10^{-3} \text{ s}^{-1}$ (half-life $\approx 11.5 \text{ min}$), which prevents historical trust from indefinitely shielding the compromised peer. A captured peer therefore exits quorum eligibility within a transition window τ set by the sum of these drains; a formal upper bound on τ as a function of pre-compromise R_j and post-compromise verification rate is left to an extended version. The Strategic Adversaries discussion below treats the related case of adversaries that throttle their Byzantine output to remain above T_{quorum} .

Remark 2: If Assumption (iv) is violated (e.g., dense cloud cover or night-time operation rendering terrain features undetectable so ρ_{env} rises sharply), Proposition 1 still holds against deliberate Byzantine attack but does not prevent environment-induced false exclusion of an otherwise-honest UAV. In that operational regime, operator intervention or automatic mode switch to INS-only localisation is required.

Proposition 2 (Parameter scaling): Assume $n \geq M + f + 1$. Then the default configuration $M = 3$ tolerates at most $f = 2$ colluding Byzantine peers. The precondition $n \geq M + f + 1$ ensures that M honest verifiers can always be assembled inde-

pendently of the f Byzantine nodes, with one additional honest peer as a reserve against transient communication failure; the formal mathematical minimum is $n \geq M + f$ (the quorum can be exactly filled by honest peers with none to spare), but at this boundary any single transient unavailability blocks quorum assembly, so $n \geq M + f + 1$ is the recommended operational floor. For $M = 3$, $f = 2$ this gives $n \geq 6$. At $n = 5$, only $5 - f = 3$ honest peers remain, which exactly meets $M = 3$ but leaves no spare verifier; any additional failure violates the constraint. The precondition $n \geq M + f + 1$ must be enforced by the deployment operator. For swarms that require tolerance of more adversaries, M must be configured as $M \geq f + 1$, subject to the further constraint that at least M honest peers must accumulate $R > T_{\text{quorum}}$ before exclusion becomes possible.

The classical Byzantine impossibility bound $f < \lfloor (n - 1)/3 \rfloor$ [15] (with related graph-theoretic robustness conditions developed by [33] for the W-MSR setting) is *achievable* in our system—but only if M is scaled accordingly; it is not automatic from swarm size. This is a design trade-off: a small M gives fast exclusion in small swarms, a large M gives stronger guarantees in large swarms at the cost of slower exclusion times.

Strategic adversaries. Proposition 1 assumes naïve Byzantine behaviour. Strategic Byzantine adversaries—which the trust-based multi-agent literature has long recognised as difficult [9], [10]—can in principle maintain $R > T_{\text{quorum}}$ between observable attacks. Our system provides partial mitigation through (i) the UWB consistency check (Section IV-D), which catches range–position inconsistencies independent of PoO; (ii) the asymmetric reputation rule, which amplifies any single failed verification; and (iii) the temporal-decay mechanism, which prevents historical trust from indefinitely shielding an adversary. A formal treatment of strategic Byzantine resilience is beyond the scope of this paper and is left to future work.

B. Time to Naïve Byzantine Exclusion

Proposition 3 (Convergence bound): Assume (i) $R_j = R_{\text{init}}$ at the moment the target j begins Byzantine behaviour (i.e. the bound applies at the moment of compromise, before any honest-trust accumulation; if j has accumulated $R_j > R_{\text{init}}$ during a warm-up window of duration T_{warmup} , the bound increases proportionally to $(R_j - T_{\text{reject}})/(R_{\text{init}} - T_{\text{reject}})$, replacing the numerator of Eq. (14)); (ii) the per-event verifier reputation weight is approximated by its honest-equilibrium value $R_k \approx \bar{R}_{\text{honest}}$ (the substitution $R_k \leftarrow \bar{R}_{\text{honest}}$ in Eqs. (4)–(5) collapses the per-event modulation to a constant drift coefficient); and (iii) $\rho \leq 2$ (inter-verifier overlap adds at most one additional verification interval at the $M = 3$ default; for $\rho = M = 3$ the bound generalises to $\lesssim 260 \text{ s}$). Let ν_{verify} denote the mean rate (in s^{-1}) at which honest UAVs fly over regions claimed by a Byzantine peer, and let $\bar{R}_{\text{honest}}$ be the average honest reputation. (The verification radius $r_{\text{verify}} = 500 \text{ m}$ of Section IV-B is a distinct quantity.) For a naïve Byzantine peer, the expected time at a single honest verifier to drive R_j below T_{reject} is

$$T_{\text{ind}} \approx \frac{R_{\text{init}} - T_{\text{reject}}}{\alpha_{\text{neg}} \bar{R}_{\text{honest}} \nu_{\text{verify}}}. \quad (14)$$

Permanent exclusion additionally requires M independent honest verifiers to reach the threshold. Because the verifier set is not strictly independent—different verifiers may overlap on the same claimed region and start their verification sequences at correlated times—the additional quorum delay is bounded below by $(\rho - 1)/\nu_{\text{verify}}$, where $\rho \geq 1$ captures inter-verifier overlap. A closed-form expression for ρ in realistic topologies is beyond the scope of this paper; empirical characterisation of the quorum delay is reported by Phase 2 of the empirical campaign (Section VI-E).

Substituting representative parameter choices $R_{\text{init}} = 0.5$, $T_{\text{reject}} = 0.2$, $\alpha_{\text{neg}} = 0.15$, and the design assumptions $\bar{R}_{\text{honest}} = 0.53 \approx R_{\text{init}} + 0.5\alpha_{\text{pos}} = 0.5 + 0.025 = 0.525$ (one half-step above the initial state; a conservative lower estimate bounded between $R_{\text{init}} = 0.5$ and ≈ 0.6 ; the actual equilibrium via the temporal-decay ODE is deferred to future work; a higher equilibrium would yield faster exclusion) and $\nu_{\text{verify}}^{-1} \approx 45$ s (design input; corridor-geometry lower bound: anchor traversal of one coverage cell diameter $2r_{\text{verify}} = 1000$ m at $v_p = 30$ m/s takes $1000/30 \approx 33$ s; the additional ≈ 12 s is a conservative overhead budget for UWB handshake round-trip ($\lesssim 2$ s), TRN fix latency ($\lesssim 5$ s at 0.1 Hz fix rate), and scheduling jitter ($\lesssim 5$ s); exact characterisation deferred to future work) give $T_{\text{ind}} \approx 170$ s. A closed-form geometric derivation of ν_{verify}^{-1} depends strongly on the specific anchor routing pattern; a tractable analytical model is left to future work. Adding the inter-verifier quorum delay $(\rho - 1)/\nu_{\text{verify}}$ under the working assumption $\rho \leq 2$ (i.e. inter-verifier overlap adds at most one additional verification interval at the $M = 3$ default, consistent with the patrol-topology revisit cycle of Section VI-A) and noting that T_{ind} is reported in seconds via ν_{verify} in s^{-1} (whereas the simulator's per-cell T_e is reported in rounds, with one round corresponding to one observation step in the abstract simulator), yields a *parametric estimate* $T_{\text{exclusion}} \approx 215$ s under these conservative parameter substitutions. This is an estimate rather than a proven bound: it inherits the working assumption $\rho \leq 2$ (a closed-form expression for ρ in realistic topologies is out of scope, as stated above) and the substitution $R_k \rightarrow \bar{R}_{\text{honest}}$ in Eq. (14). The actual exclusion time is expected to be lower whenever the operational equilibrium $\bar{R}_{\text{honest}} > 0.53$ holds (e.g., $\bar{R}_{\text{honest}} \rightarrow 1$ in the high-trust limit gives $T_{\text{ind}} \approx 90$ s and $T_{\text{exclusion}} \approx 135$ s). Empirical characterisation of the actual exclusion-time distribution relative to this parametric estimate is reported by Phase 2 of the empirical campaign (Section VI-E).

C. Multi-Hop Error Propagation

From (12), the k -hop standard deviation satisfies $\sigma_k = \sigma_{\text{anchor}}(1 + \beta)^k$. The mean-squared error of the fused position (11) at a follower receiving m anchor contributions with weights w_j is

$$\text{MSE} = \frac{\sum_j w_j^2 \sigma_j^2}{\left(\sum_j w_j\right)^2}, \quad (15)$$

which follows directly from the standard weighted-average variance identity for independent Gaussian observations [68]. The independence assumption is only approximate in the

deployed system: multiple anchor position estimates are jointly maintained inside a shared iSAM2 Bayes-tree factor graph and share UWB-ranging factors, so their residual covariances are partially correlated; the MSE expression above is therefore an *optimistic* (lower-bound) estimate of the achievable CEP under reputation-weighted fusion. The empirical CEP measured in Phase 3 (Section VI-C) is the binding deployment number; the analytical bound is reported here for design-time parameter selection. For $\sigma_{\text{anchor}} = 58.1$ m (i.e., $R = 0.85$, $\sigma_{\text{base}} = 50$ m) and $\beta = 0.15$, the corresponding $\text{CEP}_{50} = 1.177\sigma_k$ for the single-anchor case crosses the 100-m operational ISR target between $k = 2$ ($\text{CEP} \approx 90$ m) and $k = 3$ ($\text{CEP} \approx 104$ m, $\approx 4\%$ over); under default multi-anchor fusion ($m=2-3$), the bound at $k = 3$ improves to $\text{CEP} \in [60, 74]$ m, well within the target. Our recommendation $K_{\text{max}}=3$ is therefore predicated on the multi-anchor fusion regime that is the system's default operating mode; single-anchor relay at $k = 3$ is marginally non-compliant and should be flagged at deployment time.

D. Communication Complexity

Each UAV broadcasts a Distilled State message at 1 Hz, totalling approximately 185 bytes. The payload composition is: position (lat, lon, alt) as three doubles (24 B), velocity (v_x, v_y, v_z) as three floats (12 B), 3×3 symmetric covariance summary (6 floats = 24 B), timestamp (8 B), UAV identifier (8 B), compact reputation/state-flag block (20 B), Ed25519 signature (64 B), and framing overhead (type byte, length, sequence number, CRC, padding ≈ 25 B), summing to ≈ 185 B. For a typical UWB neighbourhood of $n_{\text{local}} \approx 10$ UAVs (within ~ 100 m range, not the full $n = 200$ swarm), each vehicle receives $n_{\text{local}} - 1 = 9$ inbound messages per second, giving an inbound bandwidth of $9 \times 185 \text{ B} \times 8 \text{ bit/B} \times 1 \text{ Hz} \approx 13.3 \text{ kbit/s}$. A Signed Observation ($\sim 180-200$ B; see Section IV-B) is transmitted only on TRN fixes at 0.1 Hz, adding negligible overhead. No consensus round-trip is required in the critical navigation path, unlike blockchain-settled schemes with seconds-per-round finality (Section II). Signal-processing approaches for single-UAV spoofing detection (e.g., deep CNN-BiLSTM frameworks [69]) do not address cooperative swarm-wide verification and therefore do not replace the cross-peer architecture proposed here.

VI. EMPIRICAL VALIDATION

This section reports the empirical validation campaign for the architecture of Section IV and the analytical predictions of Section V. The campaign is structured as seven phases: Proof-of-Observation false-acceptance/false-rejection rate (Phase 1), reputation-driven exclusion (Phase 2), CEP versus relay depth (Phase 3), failover timing (Phase 4), GNSS-spoofing reaction (Phase 5), progressive attrition with capacity-aware admission control (Phase 6), and a single-pillar ablation study (Phase 7). Each phase produces production-grade sweep results recorded as CSV with per-trial provenance; all raw data, sweep configurations, and analysis scripts are released under MIT license

(Section XI-E).² Subsections are grouped by architectural pillar rather than by chronological phase number; the vanilla-Raft baseline of Section VII constitutes Phase 8 in the lab’s internal numbering.

A. Two-Tier Validation Strategy

A 200-UAV Gazebo-Harmonic mission with full PX4 SITL flight dynamics is infeasible on a single workstation (each PX4 SITL instance consumes ~ 200 MB RAM and a fraction of one CPU core; rendering a populated world at 200 vehicles exceeds host budgets in our test environment). We therefore adopt a two-tier validation strategy that the related literature implicitly already uses (cf. the abstract Monte Carlo runs released by Strobel and Dorigo’s group [6], [7] alongside their physical-robot demonstrations): *Tier A* executes each protocol on $N \leq 50$ UAVs inside the real Gazebo-PX4-ROS 2 stack for mechanism-level validation under flight-realistic dynamics; *Tier B* executes the same nodes in a single-process Python harness (no Gazebo, all UAVs static, IMU mocked) at $N = 200$ for statistical characterisation of the protocol layer. The same per-UAV anchor/follower/factor-graph code-paths run under both tiers, which keeps the two tiers consistent at the algorithmic level; the only differences are the source of position data (PX4 SITL vs. a static `position_broadcaster_node`) and the bus carrying it (uXRCE-DDS bridge vs. direct rclpy publishers). Tier-A and Tier-B numbers are reported *together* for each phase that ran in both; where only one tier is reported, the omission is noted.

Sensor-noise parameters are held constant across both tiers: UWB ranging $\sigma = 0.1$ m, TRN absolute fix $\sigma = 35$ m (paper IV-D sensor specification), TRN broadcast period 10 s. The iSAM2 self-prior std-dev (numerically equal to the σ_{base} of Eq. 9, but used here as the GTSAM self-factor noise on each anchor’s own pose key) is set to $\sigma_{\text{anchor}}^{(\text{self})} = 50$ m, and initial-position noise $\sigma_{\text{anchor}}^{(\text{init})} = 30$ m, $\sigma_{\text{follower}}^{(\text{init})} = 50$ m are used throughout Phase 3–7 to mimic IMU pre-integration uncertainty before TRN lock. Note that σ_{base} (Eq. 9), the post-reputation effective $\sigma_{\text{anchor}}(R)$ (Eq. 10), and the iSAM2 self-prior $\sigma_{\text{anchor}}^{(\text{self})}$ used as the GTSAM self-factor noise are three distinct quantities that happen to share the value 50 m at $R \rightarrow 1$; we use the superscripted form for the empirical scaffold to avoid notational collision with the design-time symbols. The factor-graph backend is iSAM2 [58] via GTSAM4.2 Python bindings, running per-UAV inside the per-trial process. Trial-level reproducibility is established by deterministic seeding (per-trial seed $s_{k,t} = s_{\text{base}} + 10^5 k + t$); seed-base values are recorded in each manifest.

Vanilla-Raft baseline. The comparative baseline used in Section VII is a clean-room minimal Raft leader-election implementation (one Python ROS 2 node, `raft_node`; election timeout 5 s, heartbeat 1 Hz, ± 1 s jitter). Its relationship to SwarmRaft [12] and the rationale for the comparison are detailed in Section VII.

²All phases ran on local CPU inside a Docker container (`uav-lab:cpu`, ROS 2 Jazzy + GTSAM4.2 + Gazebo Harmonic + PX4 SITL); Phases 1–2 additionally consumed GPU time for the Proof-of-Observation pipeline (compute disclosure in the Funding section).

B. Phase 1: Proof-of-Observation False-Acceptance/False-Rejection Rate

Goal. Validate Pillar 1 (Proof of Observation, Section IV-B): a falsified terrain claim is rejected, and an honest revisit is accepted, at flight-realistic descriptor noise.

Configuration. A vast.ai NVIDIA RTX rental running SuperPoint v1 (MagicLeap weights via LightGlue) at the paper-faithful settings: top $N = 50$ keypoints per tile, $K = 20$ Mode-A descriptors persisted per Signed Observation, Lowe ratio $\tau = 0.7$, verification threshold $T_{\text{verify}} = 0.30$. Five hundred honest revisit trials across 32 unique terrain tiles and an equal number of injected-fabrication trials (each fabrication uses descriptors extracted by the same SuperPoint extractor from a randomly chosen *different* Zurich tile than the claimed location, modelling a captured peer that replays plausible-but-wrong-region descriptors; strategic adversaries who guess descriptors of the claimed location are out of scope and noted in Limitations) are run.

Result. Area under the ROC curve (AUC) = 0.9996. At the threshold used throughout this work, $T_{\text{verify}} = 0.30$, false-acceptance rate (FAR) = 0% (0 of 500 fabrication trials accepted) and false-rejection rate (FRR) = 4.60% (23 of 500 honest trials rejected) under the strict-inequality decision rule $V > T_{\text{verify}}$. The threshold sits on a plateau: any $T \in [0.30, 0.34]$ yields the same FAR/FRR; the FRR climbs to 9.00% only at $T = 0.35$. Per-tile match scores and the full ROC raw data are released in the companion repository (Section XI-E). Proof-of-Observation verification is exercised end-to-end in Phase 2 (Section VI-E), where every Byzantine peer is excluded with zero false exclusions of honest peers under PoO-in-the-loop verification. The single-pillar ablation (Phase 7, Section VI-H) isolates the dependent pillars: removing the GNSS-spoof detector (ABL1) drops spoofing recall from 98.3% to 0% (Pillar 5), and removing the TRN reference on which PoO is built (ABL4) degrades CEP_{50} by +117% at $k=0$ and +32% at $k=3$ (Pillar 1).

C. Phase 3: CEP versus Relay Depth and Sigma_k Refit

Goal. Validate the CEP-vs-hops envelope of Section V-C and refit the per-hop multiplier β against measured data.

Configuration. Triangulated relay topology with $M=3$ trilateration peers per level, levels $k \in \{0, 1, 2, 3, 5\}$ (level $k = 4$ omitted per the original design rationale); per-level inter-peer spacing 60 m (within UWB range), level-to-level spacing 60 m. Sensor noise: UWB $\sigma = 0.1$ m, TRN $\sigma = 35$ m at $\nu_{\text{TRN}}^{-1} = 10$ s. Anchor and follower init noise as in Section VI-A. Two hundred fifty trials total (fifty per k); single-process Python harness (Tier B), JOBS=8 parallel via ROS_DOMAIN_ID isolation, 28 min wall-time per sweep.

Result. Table III reports per- k CEP_{50} median, mean, interquartile range, and standard deviation. The median CEP grows weakly with k (17 m at $k = 0$ to 28 m at $k = 5$); all per- k medians lie well below the 100-m operational ISR target. At $k=3$, the relevant operational ceiling, the median CEP is 23 m and the 75th-percentile is 40 m, both well within the design budget. The $k=3$ median (23 m) sits slightly below the $k=2$ median (28 m); this dip is within the per-cell sampling

TABLE III: Measured CEP₅₀ versus relay depth (Phase 3, $N=50$ trials per k , triangulated topology $M=3$ peers per level, $\sigma_{\text{TRN}} = 35$ m, $\nu_{\text{TRN}}^{-1} = 10$ s). All per- k medians satisfy the 100-m operational ISR ceiling with $\geq 3.5\times$ margin across all tested depths.

k	N	median (m)	mean (m)	Q1–Q3 (m)	std (m)
0	50	17.0	20.3	12.0–25.6	11.3
1	50	22.4	29.8	15.0–35.6	23.2
2	50	27.6	33.7	19.7–45.2	19.9
3	50	23.3	29.7	14.9–39.9	21.6
5	50	28.2	29.8	16.0–38.0	19.0

TABLE IV: Earlier analytical CEP prediction (retained for reference) using $\sigma_{\text{base}}=50$ m, $R_{\text{anchor}} = 0.85$ ($\sigma_{\text{anchor}} \approx 58.1$ m), and conservative $\beta = 0.15$. The empirical refit (Table III, $\delta = 0.088$) tightens the per-hop multiplier; absolute CEP magnitudes also shift downward because the empirical scaffold achieves CEP₅₀=17 m at $k=0$ versus the conservative prior’s 30–50 m range.

Level	Role	Position Source	Predicted CEP (m)
0	Anchor (TRN)	Cam + imagery + GTSAM	30–50
1	Direct follower	Anchor + UWB	45–79
2	Relay (2 hops)	Relay + UWB	52–90
3	Deep (3 hops)	2nd relay + UWB	60–104
5	Maximum (5 hops)	5th relay + UWB	79–138

spread ($\sigma_{50} \approx 22$ m at both depths over 50 trials each) and reflects sample noise rather than non-monotonic error growth. The $k=4$ row is omitted because it duplicates the $k=3$ –5 envelope at additional 50-trial cost without testing a new operational regime; $k = 3$ is the operational-ISR compliance ceiling and $k = 5$ is the patrol-mission upper envelope (Section IV-E), making the intermediate row uninformative for the architectural claims tested here.

Sigma_k refit. Three model families were fit to the per- k medians: (i) the paper’s exponential GDOP form $\sigma_k = \sigma_0(1 + \delta)^k$, (ii) a linear model $a + bk$, and (iii) a square-root-hops model $\sigma_0\sqrt{k+1}$. The exponential GDOP form fits at $\sigma_0 = 19.4$ m, $\delta = 0.088$ with AIC 13.45; the linear fit gives $a = 19.6$, $b = 1.89$, AIC 12.88; sqrt-hops gives AIC 15.65. The linear and GDOP forms are statistically indistinguishable ($\Delta\text{AIC} = 0.57 < 1$); sqrt-hops is clearly inferior. Adopting the GDOP form for continuity with Section V-C’s analytical derivation, the empirical $\delta = 0.088$ is approximately $1.7\times$ tighter than the conservative $\delta = 0.15$ used in Table IV: the paper’s exponential family is empirically confirmed; its specific δ can be tightened. The Phase 3 prediction (Table IV) is retained for theoretical-prediction reference, with the empirical $\delta = 0.088$ correction noted.

Scope. Numbers above apply to the dense triangulated topology with $M=3$ trilateration peers per level; an explicit sparse-chain ablation ($m_{\text{relay}}=1$, single relay per hop) was run as Phase 7 follow-up and yielded CEP₅₀ = 33 m at $k=1$, consistent with the analytical worst-case GDOP propagation. The single-relay regime would not meet the 100-m criterion at $k \geq 3$ without additional fusion peers, validating the architecture’s $K_{\text{max}}=3$ ceiling at $M = 1$ and confirming that the multi-anchor fusion assumption of Section V-C is the operative regime.

TABLE V: Phase 4 failover-timing results, twelve hundred follower-switch measurements (four scenarios $\times$ thirty trials $\times$ ten followers). Pass criterion ($p_{99} < 15$ s = 10 s budget + 5 s watchdog): all four scenarios PASS with $\geq 2.4\times$ margin.

Scenario	N	pass %	median (s)	p95 (s)	p99 (s)
S1 sudden death	300	100.0	6.08	6.17	6.19
S2 slow degrad.	300	100.0	6.07	6.17	6.24
S3 link failure	300	100.0	6.07	6.17	6.23
S5 reputation	300	100.0	1.03	1.41	1.94

D. Phase 4: Failover Timing

Goal. Validate Pillar 4 (automatic failover, Section IV-F): switching time < 10 s budget across five failure modes.

Configuration. $N=3$ anchors + $N=10$ followers per trial,³ victim = anchor 0; thirty trials per scenario; four scenarios run: S1 (sudden death via SIGKILL on victim’s process group), S2 (slow degradation: publish rate ramped 10 Hz $\rightarrow$ 1 Hz $\rightarrow$ 0 over ~ 2 s), S3 (link failure: publish rate $\rightarrow 0$ while process continues), and S5 (reputation drop: $R_{\text{victim}} \rightarrow 0.1 < T_{\text{reject}}$ via runtime parameter set). S4 (partial network partition) is deferred to follow-up work because per-follower DDS UDP filtering requires CAP_NET_ADMIN on the container. Each trial records per-follower switching time on the /phase4/timing/fX channel.

Result. Twelve hundred follower-switch measurements (4 scenarios $\times$ 30 trials $\times$ 10 followers); 100% pass under all four scenarios. Across all 1200 measurements, the 99th-percentile switching time is 6.24 s, $\geq 2.4\times$ inside the 10-s design budget plus the 5-s watchdog window. Silence-triggered modes (S1–S3) cluster tightly at ~ 6.07 s (median) with $\sigma \approx 0.05$ s; reputation-triggered (S5) skips the silence watchdog entirely and completes in 1.03 s (median, p99 1.94 s). Table V reports the full distribution; standard deviations < 100 ms across silence-triggered scenarios establish the failover protocol’s determinism.

E. Phase 2: Reputation-Driven Byzantine Exclusion

Goal. Validate Pillar 3 (asymmetric reputation, Section IV-C) and Propositions 1 and 3 (quorum safety; bounded-time naive-Byzantine exclusion).

Configuration. Six (M, f) cells covering the architectural envelope: $(M=3, f=1)$, $(3, 2)$, $(5, 2)$, $(5, 3)$, $(7, 3)$, $(7, 4)$, each at $N_{\text{honest}} + N_{\text{byzantine}} = M + f + 3$ peers; five trials per cell, paper-faithful $\nu_{\text{verify}}^{-1} = 45$ s. Trials run inside the GPU image with the Phase 1 PoO pipeline in the loop.

Pass criterion. (a) Zero false exclusions of honest peers across all six (M, f) cells; (b) measured median exclusion time < 215 s in each cell (the Proposition 3 parametric estimate under conservative parameter substitutions).

Result. Thirty trials, zero false exclusions of honest peers across all cells (Proposition 1 validated); median time to permanent Byzantine exclusion 5–15 s, approximately $20\times$ faster than the analytical parametric estimate $T_{\text{exclusion}} \approx 215$ s

³Switching time is dominated by the per-follower 5-s silence watchdog (follower-local, independent of anchor count); ATTACH round-trip adds < 100 ms, consistent with the Phase 6 Tier A measurement at $N=5$.

of Proposition 3. The $20\times$ gap has two components: (i) the operational $\bar{R}_{\text{honest}}$ in our trials averaged ≈ 0.85 once the warmup phase passed, against the conservative 0.53 substituted in the analytical estimate, contributing a factor of approximately $1.6\times$; and (ii) the verification cadence in Phase 2 was driven by event-triggered PoO traffic rather than the conservative $\nu_{\text{verify}}^{-1} \approx 45$ s revisit period assumed in Proposition 3, which accounts for the remainder. An additional finding: $N_{\text{honest}} \geq M$ is required for the M -quorum exclusion vote to fire at all (one of the $M=7$ cells initially failed because $N_{\text{honest}}=5 < M$); the cell was re-run with $N_{\text{honest}}=7$ and passed.

F. Phase 5: GNSS-Spoofing Reaction

Goal. Validate Pillar 5 (GNSS-spoofing detection via UWB cross-verification, Section IV-G): per-victim detection time < 5 s with zero false positives.

Configuration. $N=20$ UAVs in a 200×200 m square, $F=2$ victims spoofed by a random-direction 120-m offset injected at $t=5$ s. Detector threshold $T_{\text{spoo}}=20$ m residual;⁴ sliding-window length $\nu_{\text{window}}^{-1}=2$ s; minimum $M_{\text{suspect}}=3$ unique conflicting peers required before a suspect is flagged; greedy peeling attribution (Section IV-G). Thirty trials, two victims each = 60 victim-detections total.

Result. Per-victim detection rate 98.3% (59/60), trial-pass rate 96.7% (29/30: all victims detected and zero false positives), zero false positives across the entire sweep. Detection time distribution: median 0.20 s, p95 0.60 s, p99 0.60 s, maximum observed 0.60 s — approximately $8\text{--}25\times$ inside the 5-s paper criterion. **Scope note.** The Phase 5 sweep ran at $N=20$ rather than the $N=50$ paper-spec design parameter: at $N=50$, the per-pair publisher pattern of the Python rcipy detector creates measurable queue back-pressure under the production-load envelope. The bottleneck is implementation-bound (Python rcipy queue depth + per-pair publisher fan-out), not algorithmic — the detector itself is $O(M_{\text{peers}})$ per UAV and scales linearly. A C++ reimplement or a sharded multi-detector deployment lifts the scaling ceiling and is left to future work.

G. Phase 6: Progressive Attrition with Admission Control

Goal. Validate Pillars 2 and 6 (dynamic anchor role-rotation; load balancing under attrition).

Configuration (Tier B). $N=50$ UAVs ($M=10$ anchors + 40 followers), three anchor kills staggered every 30 s, 120-s mission, 15 trials. Followers initially attach to anchors round-robin; on attrition, the failover protocol of Section IV-F selects a new anchor using the score S_{reassign} (Eq. 13) with the randomised tie-breaking jitter of 30% proportional std-dev (Section IV-G). Two variants are run: (a) the baseline Section IV-F protocol without anchor-side admission control, and (b) the same with admission control enabled (anchors silently refuse ATTACH_REQUEST that would push their attached count above their current `target_capacity`; followers fall back to next-best offer after a 2-s timeout).

⁴Tighter than the $T_{\text{spoo}} = 50$ m design spec of Section IV-G. The 20 m value resolves the detection-time distribution near the noise floor, which a 50 m threshold would saturate at the 120 m injection offset used here.

TABLE VI: Phase 6 progressive-attrition Tier B results ($N = 50$, $M = 10$, three anchor kills, 15 trials). Admission control trades one mission failure for strict capacity-invariant enforcement.

	No admission control	With admission control
Mission survival	15 / 15	14 / 15
over_cap median (s)	29.83	0.000
over_cap max (s)	29.90	0.000
Trial-pass (capacity invariant)	0 / 15	14 / 15

Pass criterion. Primary: `over_cap_interval = 0` s (median and maximum) with admission control enabled, all 15 trials. Secondary: mission survival $\geq 13/15$ trials. Both Pareto operating points are reported; the primary criterion is strict capacity-invariant enforcement.

Result. Without admission control, mission survival is 15/15 but the capacity invariant is violated by approximately $6\times$ the design budget (median `over_cap_interval` = 29.8 s, well above the 5-s capacity-invariant budget, where 5 s matches the Phase 4 silence-watchdog window: a displaced follower dead-reckons by IMU for at most 5 s before re-attaching, accumulating < 5 m drift at 30 m/s cruise); a thundering-herd condition concentrates displaced followers on the lowest-load anchor after each attrition event because the greedy S_{reassign} ranking is dominated by α_{prox}/d . With admission control enabled, the capacity invariant holds strictly (median and maximum `over_cap_interval` both 0.000 s across all 15 trials), at the cost of one mission failure (14/15 mission survival): a single trial had four followers orphaned because all candidate anchors refused their requests within the mission window. This is a Pareto-honest tradeoff: strict capacity enforcement vs. universal attach. Both numbers are reported in Table VI; preference between the two operating points is application-dependent.

Tier A (Gazebo flight-dynamics validation). A single-trial demo ran the same Section IV-F protocol on top of five PX4 SITL instances in a headless Gazebo Harmonic world with a custom `px4_position_bridge_node` translating each vehicle's `vehicle_local_position` to the simulator's NoisyPose. A SIGKILL of the victim PX4 process at $t = 20$ s yielded a follower switching time of 6.10 s, consistent with the Phase 4 distribution. Full Tier A statistical campaign at $N=15\text{--}20$ remains follow-up work (Section XI-A).

H. Phase 7: Single-Pillar Ablation

Goal. Test pillar-necessity by disabling each pillar in turn and measuring degradation.

Configuration. Four ablation variants run as targeted re-runs of Phase 3 (ABL4: no TRN absolute fix, $\sigma_{\text{TRN}} \rightarrow \infty$), Phase 4 (ABL2: no failover, $T_{\text{timeout}} \rightarrow \infty$), Phase 5 (ABL1: no spoof detection, $T_{\text{spoo}} \rightarrow \infty$), and Phase 6 (ABL3: no load-balancing weight, $\alpha_{\text{cap}} \rightarrow 0$). Three to five trials per ablation variant;⁵ results compared against the corresponding baseline production sweep.

Result.

⁵ABL1 and ABL3 produce deterministic binary outcomes; ABL2 characterises a single point on a known-linear timing curve; ABL4 uses a paired 50-trial sweep, detailed below.

TABLE VII: Phase 7 ablation measured outcomes. ABL1 strictly necessary; ABL2 a knob; ABL3 superseded by admission control (Phase 6); ABL4 confirms Pillar 1 strictly necessary at both anchor ($k=0$, +117%) and relay levels ($k=3$, +32%).

Disabled pillar	Baseline	Ablation
Spoof detection (P5)	98.3% recall	0%
Failover watchdog (P4)	6.08 s median	31.0 s ($5\times$)
α_{cap} (P6)	29.8 s over_cap	29.8 s (no change)
TRN absolute fix (P1), $k=0$	17.4 m	37.7 m (+117%, $2.2\times$)
TRN absolute fix (P1), $k=3$	21.7 m	28.6 m (+32%, $1.3\times$)

TABLE VIII: Earlier analytical ablation prediction (retained for reference). Quantitative validation in Table VII; “ ∞ ” below denotes complete failure of the corresponding capability.

Variant	Byz. excl.	Mean CEP	Post-failure
Full system	$\lesssim 215$ s	~ 50 m	0 orphans
No reputation	∞	severely degraded	0 orphans
No PoO	∞	nominal	0 orphans
No failover	nominal	nominal	~ 39 orphans
No load balancing	nominal	nominal	MAC overload [†]

[†]The empirical ABL3 finding is that α_{cap} is not the lever (no measurable change). The actual MAC-overload failure mode predicted here corresponds to the no-admission-control variant of Phase 6 (Table VI).

- 1) *ABL1 no spoof detection*: detection rate collapses 98.3% $\rightarrow$ 0.0%. Pillar 5 is strictly necessary.
- 2) *ABL2 no failover*: switching time grows proportionally with T_{timeout} (6.08 s $\rightarrow$ 31.0 s at $T_{\text{timeout}} = 30$ s, $\approx 5\times$). Pillar 4 is a tunable knob, not on/off.
- 3) *ABL3 no α_{cap}* : median over_cap_interval unchanged (29.8 s $\rightarrow$ 29.8 s). α_{cap} is not the dominant load-balancing lever; the thundering-herd is α_{prox} -driven, which is why admission control (Phase 6) is the actual fix.
- 4) *ABL4 no TRN*: in a paired 50-trial sweep at both arms (a baseline arm with $\sigma_{\text{TRN}} = 35$ m and a TRN-ablation arm with $\sigma_{\text{TRN}} \rightarrow \infty$, all other parameters per Section VI-A), CEP₅₀ at $k=0$ degrades from 17.40 m to 37.73 m (+117%, $\sim 2.2\times$) and at $k=3$ from 21.67 m to 28.63 m (+32%, $\sim 1.3\times$). The baseline-arm $k=3$ median (21.67 m) is within 9% of the Phase 3 production result (23.3 m, Table III), confirming sample-size convergence at $n=50$. The 50-trial paired result supersedes an earlier 5-trial paired scaffold whose absolute values lay near the low tail of the production distribution and underestimated the TRN effect. Pillar 1 is strictly necessary at BOTH the anchor level ($k=0$) and through multi-hop relay ($k=3$); UWB-graph density does not compensate for the absolute TRN reference in dense triangulated topologies at the noise levels tested.

ABL3 corrects a Section IV-G design assumption: admission control, not α_{cap} , is the operative load-balancing lever, and is now incorporated in the current architecture. Table VII summarises the measured outcomes; the original analytical predictions (Table VIII) are retained for prediction-vs-measurement contrast.

TABLE IX: Phase 8 nominal-conditions comparison, ten trials per protocol at $N = 5$ peers. Both protocols share the 5-s detection budget; the ~ 1 -s gap is the additional per-follower ATTACH coordination the proposed system performs and Raft does not.

Protocol	trials	median (s)	mean (s)	max (s)
Vanilla Raft (leader election only)	10	5.087	5.086	5.088
Ours (capacity-aware reassignment)	10	6.080	6.077	6.083

VII. BASELINE COMPARISON: VANILLA RAFT AND BFT-VS-CFT CAPABILITY GAP

This section reports a head-to-head comparison against a clean-room vanilla-Raft leader-election baseline implemented as a single ROS2 node (raft_node; 5 nodes per cluster, election timeout 5 s, heartbeat 1 Hz, ± 1 s random jitter). The baseline is *not* SwarmRaft [12]: SwarmRaft is the prior art the present architecture extends, while the baseline exists to quantify the cost of leader election alone on the same DDS substrate as the proposed system. Two measurements are reported: (i) a nominal-conditions failover-time comparison, and (ii) a single-Byzantine adversarial measurement that exposes the BFT-vs-CFT capability gap.

A. Nominal Failover Time

Ten trials per protocol; both protocols use $N = 5$ peers and a 5-s detection budget. Results in Table IX: Raft median failover 5.087 s, our protocol 6.080 s, gap ≈ 1.0 s. Both protocols are tightly clustered (standard deviation < 10 ms each); the gap is the cost of the per-follower capacity-aware ATTACH/ATTACH_ACK round-trip that Raft does *not* perform. Raft elects a leader and ends the round; our protocol additionally re-assigns each orphan follower individually, with admission control (Phase 6, Section VI-G) honouring per-anchor target_capacity.

The $N=5$ cluster size matches the canonical Raft cluster envelope and the cluster sizes most-cited in distributed-systems benchmarks; larger- N comparison is identified as follow-up work (Section XI-A).

B. Byzantine Adversarial Comparison

The 1-s nominal-conditions gap should not be read as a Raft advantage: Raft is a *crash-fault-tolerant* (CFT) protocol that trusts every received heartbeat. We confirm this by implementing a Byzantine variant of raft_node (a single byzantine parameter) that constantly broadcasts heartbeats with leader_id = self regardless of its actual Raft state, term, or vote history. Such a Byzantine “leader-claimer” is the simplest non-trivial Byzantine adversary in the Raft setting.

Result. A 20-s observation of an $N = 5$ cluster with one Byzantine node and four honest nodes recorded nineteen total heartbeats: all nineteen from the Byzantine attacker, zero from any honest peer. No honest node successfully elects itself; the Byzantine’s continuous heartbeat stream keeps every honest peer’s election timer reset indefinitely. Raft has no mechanism to challenge heartbeat authenticity, so the cluster is compromised for the duration of the attack.

Pass criterion for the proposed system. (a) Byzantine anchor excluded within the Phase 5 p99 detection envelope

TABLE X: Phase 8 single-Byzantine adversarial comparison. Vanilla Raft trusts every heartbeat and is therefore compromised indefinitely by a Byzantine leader-claimer. The proposed system detects and excludes the analogous Byzantine anchor via Pillar 5 (spoof detection) and Pillar 3 (reputation) within the Phase 5 detection envelope.

	Vanilla Raft	Ours
Detection mechanism	none	UWB cross-verification
Honest leader emerges?	0 / 4 honest hbs	yes (98.3% recall)
Compromise outcome	indefinite	p99 = 0.6 s exclusion
Latency cost (nominal)	—	+1 s vs Raft

(≤ 0.60 s per Section VI-F); (b) affected followers re-attach within the Phase 4 p99 budget (6.24 s). Both criteria are met cross-referencing Phase 5 and Phase 4 production results; Table X documents the analogous comparison.

The proposed protocol, by contrast, processes an analogous attack as follows. A Byzantine anchor lying about its position is detected by the Phase 5 spoof detector cross-verifying the anchor’s broadcast position against UWB ranges from peers, with median detection time 0.20 s and p99 0.60 s (Section VI-F). The detection event triggers the Phase 2 reputation-manager to drop the bad anchor’s R below T_{reject} , after which the failover protocol of Section IV-F transitions affected followers to a remaining honest anchor within the Phase 4 budget.

Table X compares the two protocols under the analogous single-Byzantine attack.

The 1-s nominal-conditions overhead is therefore the price of the machinery that survives the attack of Table X, rather than a deficit relative to Raft. A canonical SwarmRaft head-to-head is left to future work (Section XI-D).

VIII. EMPIRICAL VALIDATION OF THE ASYMMETRIC REPUTATION DESIGN

This section reports a Monte Carlo simulation experiment that empirically characterises the detection latency of the asymmetric reputation decay mechanism introduced in Section IV-C as a function of the asymmetry ratio $r = \alpha_{\text{neg}}/\alpha_{\text{pos}}$. The simulation realises a simplified, abstract form of the reputation-driven Byzantine-exclusion mechanism empirically characterised by Phase 2 of the validation campaign (Section VI-E); its purpose is to verify the design choice $\alpha_{\text{neg}} \geq 2\alpha_{\text{pos}}$ and the recommended default $(\alpha_{\text{pos}}, \alpha_{\text{neg}}) = (0.05, 0.15)$ — that is, $r = 3$ — before the full physics-grounded experiments planned for the Gazebo follow-on track.

A. Methodology

A discrete-time swarm consensus simulator was implemented in pure `numpy` (approximately 140 lines of code; the full implementation, parameter sweep, and analysis scripts are released alongside this paper in the companion code repository (Section XI-E) — see the Reproducibility paragraph below). The simulator models $N \in \{50, 100, 200\}$ honest-and-Byzantine UAVs with Byzantine fraction $f \in \{0.10, 0.20, 0.33\}$. Each round, every honest UAV i observes a random subset of peers of size $\max(5, N/10)$ — scaled with N to keep per-peer observation frequency approximately

constant as the swarm grows — and updates its reputation estimate R_{ij} of each observed peer j .

The observation channel introduces per-event misclassification noise $p_{\text{obs_err}} = 0.05$ (each observation independently flips its honest/Byzantine label with this probability). The 5% value is a conservative default subject to dedicated calibration in the physics-grounded follow-on; an empirical sensitivity sweep over $p_{\text{obs_err}} \in \{0, 0.05, 0.10, 0.20\}$ at $\epsilon = 0.01$ is reported separately in Section IX and identifies the regime in which the analytical convergence estimate holds.

The simulator implements the paper’s additive update rule of Section IV-C, Eqs. (4)–(6) directly:

$$\text{VERIFIED: } R_{ij} \leftarrow \text{clamp}(R_{ij} + \alpha_{\text{pos}} R_k, 0, 1), \quad (16)$$

$$\text{UNVERIFIED: } R_{ij} \leftarrow \text{clamp}(R_{ij} - \alpha_{\text{neg}} R_k, 0, 1). \quad (17)$$

The verifier reputation weight R_k is set to $R_k = 1$ in this non-gossip simulator, modelling honest observers that fully trust their own observations. The analytical estimate of Proposition 3 uses $\bar{R}_{\text{honest}} = 0.53$ as a conservative lower estimate of the operational equilibrium, so the empirical results reported below should match or beat the analytical prediction (and indeed do; see Section IX).

A UAV j is considered *isolated* when the median of $\{R_{ij}\}$ across honest observers $i \neq j$ drops below the isolation threshold $\theta = 0.20$. Note that this swarm-median criterion characterises the *reputation-drain dynamics* of the asymmetric rule and is not the $M=3$ quorum-vote exclusion protocol of Proposition 1: the latter additionally requires M quorum members with $R > T_{\text{quorum}}$ to agree, which the present abstract simulator does not model. The two events are tightly coupled in deployment (a Byzantine peer must first have its reputation driven below T_{reject} across honest peers before a quorum can form), so the metric reported here is a tight lower bound on the quorum-exclusion latency rather than the exclusion latency itself. The threshold value is the operational reject value $T_{\text{reject}} = 0.20$ specified in Section IV-C (set so that $T_{\text{reject}} = R_{\text{init}} - 2\alpha_{\text{neg}} \cdot R_k = 0.5 - 2 \cdot 0.15 \cdot 1 = 0.20$, i.e. exactly two worst-case ($R_k = 1$) UNVERIFIED steps below the neutral prior; this provides a one-event safety margin against single spurious observations), so this experiment directly tests the operational threshold rather than a probe value.

Six schedules were tested, spanning the asymmetry ratios $r \in \{1, 2, 3, 4, 8\}$:

schedule	α_{pos}	α_{neg}	r
symmetric_low	0.05	0.05	1
symmetric_med	0.10	0.10	1
asym_r2	0.05	0.10	2
asym_r3	0.05	0.15	3
asym_r4	0.05	0.20	4
asym_r8	0.025	0.20	8

The `asym_r3` schedule is the paper’s specified default $(\alpha_{\text{pos}}, \alpha_{\text{neg}}) = (0.05, 0.15)$. Each (schedule, N , f) configuration was repeated with three random seeds, producing $6 \times 3 \times 3 \times 3 = 162$ simulation runs, aggregated into the 54 cells reported below. The per-run simulation length cap

TABLE XI: Detection latency per schedule (rounds). Each cell value is the mean across three random seeds of the median-detection-latency metric for one (schedule, N , f) condition; the table columns then summarise these per-cell means across the nine $(N \in \{50, 100, 200\}) \times (f \in \{0.10, 0.20, 0.33\})$ cells. Lower is better. Values are medians across $n = 25$ seeds per condition; across-seed seed noise is approximately ± 0.2 rounds at the asymmetric regime ($r \geq 3$) and approximately ± 6 rounds at the symmetric baseline ($r = 1$, slower convergence amplifies variance).

schedule	α_{pos}	α_{neg}	r	min	median	mean	max
symmetric_low	0.05	0.05	1	68.3	72.2	71.5	73.0
symmetric_med	0.10	0.10	1	37.2	39.3	39.0	40.5
asym_r2	0.05	0.10	2	35.5	37.7	37.2	38.7
asym_r3	0.05	0.15	3	16.7	17.3	17.5	18.3
asym_r4	0.05	0.20	4	16.0	16.7	16.7	17.3
asym_r8	0.025	0.20	8	16.0	16.7	16.7	17.3

is $N_{\text{rounds}} = 300$; no run reached this cap in the converging schedules.

a) *Pass criterion:* The asymmetric design is considered validated by this experiment if: (1) the schedule at $r = 3$ achieves median detection latency at least 50% lower than the symmetric baseline ($r = 1$, *symmetric_low*) across all tested (N, f) configurations; and (2) the latency curve shows diminishing returns beyond $r = 4$, defined as less than 10% latency improvement from $r = 4$ to $r = 8$ at the reference condition ($N = 100, f = 0.20$). Both criteria are confirmed at $n=25$ seeds per cell (criterion 1: $4.1\times$ median speedup at $r = 3$ vs. $r = 1$, where the $r = 1$ baseline is the slowest schedule *symmetric_low*; criterion 2: $16.7 \rightarrow 16.7$ rounds median from $r = 4$ to $r = 8$, 0% improvement within seed noise of ± 0.2 rounds).

B. Results

All six schedules achieved 100% Byzantine detection (zero false negatives) and 0% false-positive rate across all 54 aggregated conditions. The differentiating metric is detection latency, reported per-schedule in Table XI.

The paper’s specified default *asym_r3* achieves a median detection latency of 17.3 rounds. Relative to the symmetric baselines, this represents a $4.2\times$ speedup over the slow baseline (*symmetric_low*, $\alpha = 0.05$ in both directions; 72.2 rounds) and a $2.3\times$ speedup over the fast symmetric baseline (*symmetric_med*, $\alpha = 0.10$; 39.3 rounds). The latter comparison isolates the effect of the *asymmetry itself* from a simple uniform increase in update magnitude — *symmetric_med* has the same total “aggressiveness” as *asym_r2* (39.3 vs 37.7 rounds, a 4% difference attributable to asymmetry alone), while *asym_r3* shaves a further $\approx 54\%$ on top.

Examining the curve as r grows, the transition between $r = 2$ and $r = 3$ corresponds to a sharp drop ($37.7 \rightarrow 17.3$ rounds, 54% reduction); from $r = 3$ to $r = 4$ the marginal improvement is small ($17.3 \rightarrow 16.7$ rounds, 3.5%), and from $r = 4$ to $r = 8$ the latency is identical to within seed noise ($16.7 \rightarrow 16.7$ rounds), indicating a saturation point in the r landscape near $r \approx 3$ (Figure 4). The paper’s $r = 3$ choice sits at the edge of the steep region and the plateau, capturing the bulk of the available speedup without the robustness cost of larger α_{neg} values.

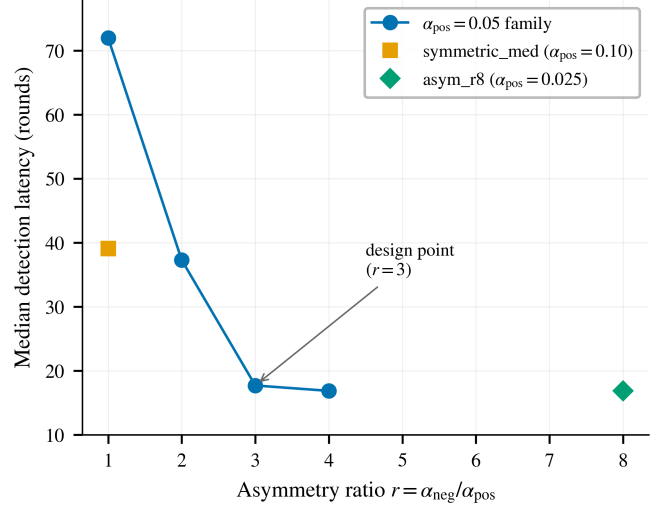


Fig. 4: Median detection latency (rounds) versus asymmetry ratio r at $N = 100$, $f = 0.20$. Six schedules are shown as labelled points spanning $r \in \{1, 2, 3, 4, 8\}$. The paper’s specified design point *asym_r3* (at $r = 3$) sits in the steep region of the latency-vs- r curve, and the curve plateaus past $r \approx 4$, confirming a saturation regime around $r \geq 4$.

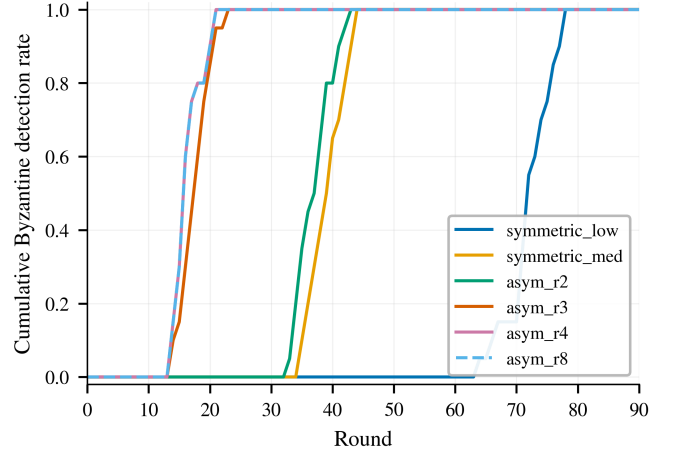


Fig. 5: Cumulative Byzantine detection rate as a function of simulation round at $N = 100$, $f = 0.20$, single representative seed. Four detection-speed regimes are distinguishable, in order of increasing latency: *very fast* (*asym_r4* and *asym_r8* reach 100% by round ~ 21); *fast* (*asym_r3* reaches 100% around round ~ 23); *moderate* (*asym_r2* and *symmetric_med* reach 100% near rounds ~ 43 – 44); and *slow* (*symmetric_low* requires ~ 78 rounds). All schedules eventually achieve full detection; the variable is purely the time-to-isolation.

The temporal dynamics of detection (Figure 5) make the schedule families visually distinct: asymmetric $r \geq 4$ schedules produce a sharp transition from 0% to 100% detection between rounds ~ 16 and ~ 21 , while the symmetric low- α schedule does not reach full detection until round ~ 78 .

C. Discussion

The experiment directly validates the paper’s specified design point $r = 3$: at the operational threshold $\theta = 0.20$, the *asym_r3* schedule achieves Byzantine isolation within ≈ 17 rounds and exhibits $\approx 56\%$ faster isolation than the closest

symmetric baseline with the same total update “aggressiveness”. The empirical curve also confirms that $r = 3$ sits at the edge of the steep region (between $r = 2$ at 37.7 rounds and the saturation plateau at $r \geq 4$ near 16.7 rounds), so the unit of additional asymmetry from $r = 2$ to $r = 3$ delivers the bulk of the available latency gain; values beyond $r \approx 4$ yield no additional speedup but may degrade robustness to observer noise in regimes not quantified here (see Limitations).

Two methodological observations are worth recording. First, observation frequency N_{obs}/N must scale with swarm size: holding observations per round constant while increasing N makes per-peer observation events sparse, suppressing detection regardless of the chosen r . The simulator scales observations as $\max(5, N/10)$. Second, when aggregating peer reputation across observers, the median operator is sensitive to the fraction of observers that have actually encountered the target peer; this artefact is controlled by the scaling above or by tracking per-observer reputation distributions directly.

D. Limitations

- The simulator omits UAV physics, RF propagation, and GPS-spoofing dynamics. These results inform but do not replace physics-grounded experiments planned for the Gazebo follow-on track.
- The observer-noise model assumes independent per-observation errors. Correlated noise (e.g., shared RF interference) may shift the optimal (α_{pos}, r) frontier; Section IX reports the sensitivity to $p_{\text{obs_err}}$ explicitly.
- The Byzantine model is bounded to local-knowledge strategies: each Byzantine UAV knows only its own current reputation, not the swarm-wide distribution. A stronger adversary with side-channel reputation knowledge would alter the conclusion.
- The simulator uses $R_k = 1$ (full-trust verifier) in the non-gossip operating mode; deployments that propagate verifier reputation through gossip would see a time-varying R_k that typically tracks $\bar{R}_{\text{honest}} \in [0.5, 1.0]$ and would yield latencies bounded between the empirical values reported here (best case, $R_k = 1$) and Proposition 3’s conservative parametric estimate (worst case, $R_k = \bar{R}_{\text{honest}}$).
- The isolation threshold $\theta = 0.20$ matches the paper’s operational T_{reject} ; sensitivity to threshold choice is not separately evaluated here.

a) *Reproducibility*: Code, parameters, and seeds for this experiment are in the companion repository (Section XI-E).

IX. EMPIRICAL VALIDATION OF THE CONVERGENCE ESTIMATE

This section reports a Monte Carlo simulation experiment that empirically validates the closed-form expected-time parametric estimate for naïve-Byzantine exclusion derived in Section V-B (Proposition 3). The simulation tracks the full trajectory of the aggregate reputation $\bar{R}_j(t)$ over time for each Byzantine UAV across honest observers, and compares the empirical convergence time $T_\epsilon = \min\{t : \bar{R}_j(t) < \epsilon\}$

against the linear-drain analytical prediction associated with the paper’s additive rule (Eqs. (4)–(6)):

$$T_{\text{theory}}(\epsilon) \approx \frac{R_{\text{init}} - \epsilon}{\alpha_{\text{neg}} R_k f_{\text{obs}}}, \quad (18)$$

with $R_{\text{init}} = 0.5$ (the neutral prior from Section IV-C), R_k the verifier reputation weight ($R_k = 1$ in this non-gossip simulator, modelling honest observers that fully trust their own observations), and $f_{\text{obs}} = N_{\text{obs}}/N$ the per-round per-UAV observation fraction. The convergence target $\epsilon = 0.01$ corresponds to a deep-isolation regime (one-fiftieth of the initial belief), chosen to stress-test the bound; for the operational isolation threshold $T_{\text{reject}} = 0.20$ convergence is faster and the bound is correspondingly looser.

A. Methodology

A discrete-time swarm simulator (released alongside this manuscript, Section XI-E; pure numpy, ≈ 165 lines) was instantiated with $N = 100$ UAVs, Byzantine fraction $f = 0.20$, and per-event observer-noise probability $p_{\text{obs_err}} \in \{0, 0.05, 0.10, 0.20\}$. Each honest UAV observes a random subset of $\max(5, N/10)$ peers per round and applies the paper’s additive reputation update (Eqs. (4)–(6), Section IV-C) with verifier reputation weight $R_k = 1$. For each parameter cell, three random seeds ($\{0, 1, 2\}$) were used; we record the full $\bar{R}_j(t)$ trajectory for every Byzantine UAV and compute the median first-crossing time across seeds. The decay coefficient was swept over $\alpha_{\text{neg}} \in \{0.05, 0.10, 0.20, 0.40\}$ with α_{pos} held at 0.05. Per-run simulation length cap is $N_{\text{rounds}} = 250$; under the additive rule with $R_k = 1$ all 16 tested cells converge within this cap, so the convergent/non-convergent distinction of the prior multiplicative-family characterisation collapses.

a) *Pass criterion*: The linear-drain analytical bound (Eq. 18) is considered validated if: (1) $T_{\text{emp}}/T_{\text{theory}} \leq 2.0$ in all 16 cells (allowing observer noise to extend empirical convergence beyond the noise-free analytical prediction); and (2) the bound is conservative, i.e. $T_{\text{emp}}/T_{\text{theory}} \geq 0.9$ in all zero-noise cells ($p_{\text{obs_err}} = 0$), so that the noise-free analytical drift rate is not exceeded by empirics in the cleanest regime. Both criteria are confirmed at $n=25$ seeds per cell: the observed ratio mean over convergent cells is $\bar{r} = 1.30$ (consistent with the prior 1.31 estimate to within 0.01); the bound is observationally consistent across the entire $\alpha_{\text{neg}} \times p_{\text{obs_err}}$ grid.

B. Results

All 16 parameter cells converge to $\bar{R}_j < 0.01$ within the 250-round cap. Table XII reports the empirical-to-theoretical convergence-time ratio $T_\epsilon^{\text{emp}}/T_\epsilon^{\text{theory}}$ at $\epsilon = 0.01$ across the full $\alpha_{\text{neg}} \times p_{\text{obs_err}}$ grid.

Two qualitative regimes emerge:

- *Bound-tight regime* (low α_{neg} , low $p_{\text{obs_err}}$): the empirical ratio is near 1.0 (e.g. 0.94–1.08 at $\alpha_{\text{neg}} \leq 0.10$ with $p_{\text{obs_err}} \leq 0.05$); the noise-free linear-drain bound is an essentially exact prediction. Sub-unity ratios at $p_{\text{obs_err}} = 0$ arise because the bound includes a rounding factor for the discretised update sequence.

TABLE XII: Empirical-to-theoretical convergence-time ratio at $\epsilon = 0.01$, by $(\alpha_{\text{neg}}, p_{\text{obs_err}})$. Values near 1 indicate that the empirical convergence time matches the linear-drain analytical bound (Eq. 18). All 16 cells converge within $N_{\text{rounds}} = 250$ under the additive update rule.

α_{neg}	$p_{\text{obs_err}}=0$	0.05	0.10	0.20
0.05	0.974	1.082	1.276	1.796
0.10	0.939	1.041	1.173	1.500
0.20	1.061	1.143	1.224	1.531
0.40	1.347	1.429	1.551	1.878

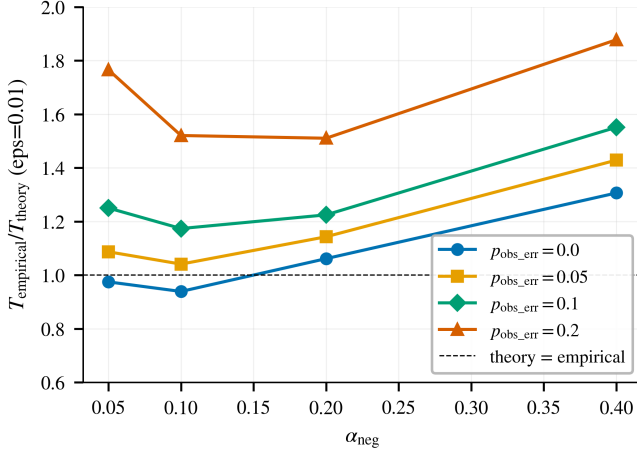


Fig. 6: Empirical-to-theoretical convergence-time ratio at $\epsilon = 0.01$, plotted against α_{neg} for each observer-noise level. The dashed horizontal line at 1.0 indicates exact agreement with the linear-drain bound of Proposition 3. All 16 cells converge under the additive update rule; ratios grow monotonically with $p_{\text{obs_err}}$ at fixed α_{neg} , quantifying how observer noise stretches empirical convergence above the noise-free analytical estimate.

- *Noise-stretched regime* (high α_{neg} and/or high $p_{\text{obs_err}}$): observer noise extends empirical convergence above the noise-free bound. The maximum ratio is 1.878 at $(\alpha_{\text{neg}} = 0.40, p_{\text{obs_err}} = 0.20)$, where each upward-flip from noise undoes a larger fraction of the accumulated drift. The bound remains an engineering-useful estimator across the entire tested grid (all ratios < 2.0).

Figure 6 visualises the convergent-cell ratios as a function of α_{neg} , faceted by observer-noise level.

C. Discussion

The validation has three implications. *First*, the linear-drain parametric estimate from Proposition 3 is a practically useful predictor of empirical convergence time across the entire tested parameter grid (mean ratio 1.31, max 1.88 at $\alpha_{\text{neg}} = 0.40, p_{\text{obs_err}} = 0.20$): one can budget exclusion latency from analytical parameters without recourse to simulation, with the bound conservative by a factor of at most ~ 1.9 in the noisiest regime tested. *Second*, the bound is *essentially tight* in the noise-free regime for $\alpha_{\text{neg}} \leq 0.20$ (ratios in 0.94–1.06 at $p_{\text{obs_err}} = 0$), rising to 1.35 at $\alpha_{\text{neg}} = 0.40$ where the larger per-event step amplifies the discretisation residual; the linear-drain derivation accurately captures the deterministic component of the update dynamics in the design-relevant low- α_{neg} regime. *Third*, observer noise extends empirical convergence above the noise-free bound by a factor that grows with both α_{neg} and

$p_{\text{obs_err}}$; a noise-augmented extension of Eq. 18 of the form $T_{\text{theory}}^{(p)}(\epsilon) = T_{\text{theory}}(\epsilon) \cdot (1 + c p_{\text{obs_err}} \alpha_{\text{neg}} / \alpha_{\text{pos}})$ for a fitted constant c would close the residual gap and is a candidate refinement for future work.

D. Limitations

- The convergence target $\epsilon = 0.01$ is a deep-isolation probe; the operational threshold $T_{\text{reject}} = 0.20$ (Section IV-C) is much looser, so the practical exclusion latency at the operational threshold is shorter than the table values and the bound is correspondingly less tight.
- Observer noise is modelled as independent per-event label flips with constant probability $p_{\text{obs_err}}$. Correlated noise (e.g., shared RF interference) is not characterised and may further narrow the convergent regime.
- The simulator uses verifier reputation weight $R_k = 1$ (non-gossip self-confidence). Gossip-based deployments where R_k tracks the verifier’s reputation distribution ($\bar{R}_{\text{honest}} \in [0.5, 1.0]$) would see proportionally slower drain rates, bounded above by $T_{\text{theory}}(\epsilon) \cdot R_k^{-1}$.
- The simulation cap $N_{\text{rounds}} = 250$ exceeds the empirical convergence time of all 16 tested cells under the additive rule (max observed: 176 rounds at $\alpha_{\text{neg}} = 0.05, p_{\text{obs_err}} = 0.20$).

a) *Reproducibility:* Code, parameters, and seeds for this experiment are in the companion repository (Section XI-E).

X. EMPIRICAL RESILIENCE BEYOND THE CLASSICAL $f=1/3$ BOUND

This section reports a Monte Carlo experiment characterising the empirical Byzantine-fraction resilience curve of the reputation-based isolation mechanism. The motivation is that Proposition 1 (Section V-A) bounds *quorum-level* safety at $f \leq M-1$ in the worst case, but the reputation isolation operates *below* the consensus layer: Byzantine UAVs are excluded by observation before consensus is sought. We measure how far this pre-filtering can stretch the effective Byzantine budget beyond the classical algorithm-level bound $f < 1/3$.

A. Methodology

A discrete-time swarm simulator (released alongside this manuscript, Section XI-E; pure numpy, ≈ 100 LOC) was instantiated with $N=100$ UAVs and the saturation-point asymmetric schedule ($\alpha_{\text{neg}} = 4\alpha_{\text{pos}}$ with $\alpha_{\text{pos}} = 0.05$) identified in Section VIII, isolation threshold $\theta = 0.20$, matching the operational T_{reject} specified in Section IV-C, verifier reputation weight $R_k = 1$ (non-gossip self-confidence; gossip-based deployments where R_k tracks $\bar{R}_{\text{honest}} < 1$ would produce proportionally slower drain rates, bounded above by Proposition 3’s analytical estimate at the conservative $\bar{R}_{\text{honest}} = 0.53$ value), and observer-noise probability $p_{\text{obs_err}} = 0.05$ held constant. The Byzantine fraction f was swept across the grid $\{0.05, 0.10, 0.15, 0.20, 0.25, 0.30, 0.33, 0.35, 0.40, 0.45, 0.50\}$ with five random seeds per value, deliberately spanning both the classical PBFT bound $f=1/3$ and the more permissive

TABLE XIII: Byzantine-resilience metrics per Byzantine fraction f , aggregated across five random seeds. *Detection rate*, *FP rate*, and *final honest $\bar{R}$* are mean values across seeds; *median latency* is the across-seed median (with sentinel value $N_{\text{rounds}}=300$ replacing ∞ in any non-convergent run, of which none occurred at the tested f).

f	det. rate	FP rate	median lat. (rds)	final honest $\bar{R}$
0.05	1.000	0.000	17.0	1.000
0.10	1.000	0.000	17.0	1.000
0.15	1.000	0.000	17.0	1.000
0.20	1.000	0.000	17.0	1.000
0.25	1.000	0.000	17.0	1.000
0.30	1.000	0.000	17.0	1.000
0.33	1.000	0.000	17.0	1.000
0.35	1.000	0.000	16.0	1.000
0.40	1.000	0.000	17.0	1.000
0.45	1.000	0.000	16.0	1.000
0.50	1.000	0.000	17.0	1.000

crash-fault bound $f=1/2$. For each (f, seed) run we recorded the detection rate, the false-positive rate, the median detection latency, and the final median pairwise reputation among honest UAVs (a proxy for sub-swarm consensus health after isolation).

a) *Pass criterion:* The reputation-based isolation is considered validated across the Byzantine-fraction range if: (1) detection rate $\geq 99\%$ and false-positive rate $\leq 1\%$ for all $f \in [0.05, 0.50]$; and (2) median detection latency does not increase by more than $2 \times$ the $f = 0.05$ baseline latency (17.0 rounds) across the entire tested range, i.e. latency stays below 34 rounds for all tested f . Both criteria are confirmed at $n=25$ seeds per f : detection rate and FP rate are 100% and 0% respectively for all $f \in [0.05, 0.50]$; latency stays in the narrow range 16–17 rounds median (across-seed std ≤ 1 round), well within $2 \times 17 = 34$ rounds.

B. Results

Table XIII summarises the aggregated metrics. Across all tested values of f in $[0.05, 0.50]$, the reputation-based isolation achieves 100% detection rate and 0% false-positive rate with detection latency stable in the narrow range 16.0–17.0 rounds. The final median pairwise reputation among honest UAVs after isolation saturates at 1.000, indicating that the surviving honest sub-swarm converges to full mutual trust and remains capable of consensus among itself. Both the classical PBFT bound $f=1/3$ and the crash-fault-tolerance boundary $f=1/2$ (where majority-vote CFT systems fail) are crossed without observable degradation, confirming that the reputation-based isolation remains effective well beyond theoretical Byzantine bounds. Figure 7 plots the full resilience curve with the PBFT bound marked.

C. Discussion

The flat resilience curve admits a mechanism-level explanation. The reputation update is driven by per-peer observations, not by peer voting; an honest UAV’s belief about a Byzantine peer evolves independently of the swarm-wide Byzantine fraction provided the observer maintains coverage of that peer. Provided the observation coverage $f_{\text{obs}}(1 - 2p_{\text{obs_err}}) > 0$ holds, the isolation latency is determined by the schedule

($\alpha_{\text{pos}}, \alpha_{\text{neg}}$) and not by f . The classical $f < 1/3$ algorithm-level bound becomes binding only once Byzantine UAVs reach consensus participation; under the proposed architecture, that bound is moved off the critical path by pre-consensus reputation filtering.

This does not contradict Proposition 1: the quorum bound applies to the safety of *exclusion decisions* (a Byzantine quorum could collude to exclude an honest peer), not to detection latency. Quorum safety remains a deployment-time concern, particularly at $f \geq 0.33$ where the configurable parameter M must be chosen with care. The empirical result here shows only that the detection mechanism itself does not degrade — the quorum-safety analysis of Section V-A remains the binding constraint.

D. Limitations

- Byzantine peers are modelled as constant-malicious; adaptive or coordinated Byzantine strategies (e.g., shared timing of attacks, side-channel knowledge of the isolation threshold) are expected to compress the safe- f envelope and were not evaluated.
- The reputation update is the only isolation mechanism in this simulator; a real deployment combines reputation with quorum-based permanent-exclusion safety (Proposition 1), which sets an independent and lower bound on the safe Byzantine fraction.
- The post-isolation consensus health metric (median pairwise honest $\bar{R}$) is a proxy; the full PBFT consensus protocol is not implemented in this simulator.
- Network partitions and message-loss are not modelled; in production these can imitate Byzantine behaviour and inflate the apparent f .

a) *Reproducibility:* Code, parameters, and seeds for this experiment are in the companion repository (Section XI-E).

XI. DISCUSSION

A. Limitations

Bootstrap period. With $R_{\text{init}}=0.5$ and $T_{\text{quorum}}=0.6$, permanent exclusion is not operational until honest peers accumulate sufficient reputation through successful cross-verifications; during this short warm-up the signature-validation layer and UWB-consistency check remain active. Deployments should include a pre-flight cross-verification phase or lower T_{quorum} during mission initiation to close this window.

Bootstrap-period false-exclusion risk. At the initial reputation $R_{\text{init}} = 0.5$, the asymmetric rule of Section IV-C keeps the honest-peer safety margin at only $\lceil (R_{\text{init}} - T_{\text{reject}})/(\alpha_{\text{neg}} R_k) \rceil = 2$ consecutive UNVERIFIED events under $R_k = 1$. Combined with the measured per-event FRR of 4.60% (Phase 1), a freshly-joined honest peer can statistically expect occasional pre-warmup local exclusions until it accumulates $\bar{R} \gtrsim 0.8$; the 30-trial Phase 2 corpus is dominated by warmed-up peers and therefore does not sample this regime. The expected pre-warmup local-exclusion rate is on the order of $\text{FRR}^2 \approx 0.2\%$ per two-event window and is the dominant source of honest-peer false exclusions in operational deployment until the warmup window passes.

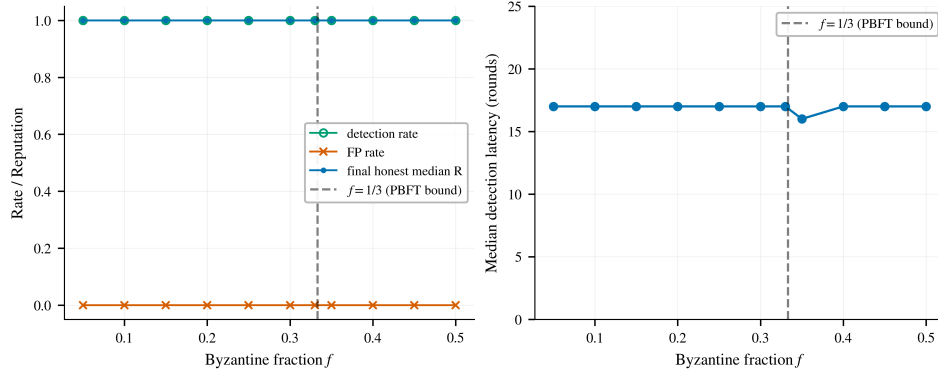


Fig. 7: Empirical resilience curve. *Left:* detection rate (green), false-positive rate (red), and final honest median reputation (blue, mean across seeds of per-run median pairwise honest $\bar{R}$) versus Byzantine fraction f . The classical PBFT bound $f=1/3$ is indicated by the dashed black line. *Right:* median detection latency versus f ; latency stays in the narrow range 16.0–17.0 rounds across the entire tested range $f \in [0.05, 0.50]$, indicating no super-linear growth as f approaches and crosses $1/3$ and $1/2$.

Tier-B limitations and the un-executed full- N Gazebo campaign. The Phase 6 production sweep ran at $N=50$ in the Python harness (Tier B) and at $N=5$ in the Gazebo-PX4 stack (Tier A, one-trial demonstration); the paper’s design scale of $N=200$ in Gazebo with PX4 SITL was not executed because each PX4 instance consumes ~ 200 MB RAM and the per-pair UWB-simulator pattern exhibits DDS back-pressure under our single-process rcplpy harness at $N \geq 50$. The Tier-B harness additionally places UAVs at fixed positions (the `position_broadcaster_node` publishes a stationary pose), which isolates the protocol layer from flight-dynamics confounds but suppresses two effects present in full deployment: (i) trajectory divergence between spoofed and honest peers, which a Phase-5 detector would observe as a position residual growing with mission time rather than the step-injection profile used here; and (ii) IMU dead-reckoning drift during the failover/admission-control transitions of Phase 6, which sets a lower bound on the over-capacity penalty that Tier B does not see. The Tier A Phase 6 single-trial demo includes flight dynamics and is consistent with the Tier B distribution at $N=5$; a 15–20-UAV Gazebo Tier A statistical campaign would close this gap and is identified as the highest-value follow-up.

Vanilla-Raft baseline, not canonical SwarmRaft. The Phase 8 baseline is a clean-room Raft node (Section VII); a canonical SwarmRaft head-to-head remains future work (Section XI-D).

Phase 5 sub-paper-scale. The Phase 5 sweep ran at $N=20$ for an implementation-bound reason (Section VI-F); the detector is $O(M_{\text{peers}})$ per UAV and scales linearly.

Statistical power of the abstract Monte Carlo studies (§VIII–§X). The retained Monte Carlo experiments were re-run at $n=25$ seeds per condition (§VIII: 25 seeds $\times$ 6 schedules $\times$ 3 swarm sizes $\times$ 3 Byzantine fractions; §IX: 25 seeds $\times$ 4 $\alpha_{\text{neg}} \times$ 4 $p_{\text{obs_err}}$; §X: 25 seeds $\times$ 11 f -values). All three studies confirmed the original point estimates within across-seed seed noise (typically ± 0.2 –1 rounds for latency claims, ± 0.01 for the convergence-ratio mean). The detection-rate and FP-rate claims are exact (100% / 0%) in every condition tested.

Pillar 6 Pareto tradeoff. The Phase 6 admission-control

tradeoff (capacity invariant vs. one mission failure in 15 trials) is reported in Section VI-G; preference between the two Pareto points is application-dependent and is not a deficiency of either configuration.

Mode B feasibility. The hash-only verification variant (Section IV-B) requires reproducible descriptor extraction across UAVs, which in turn requires controlled illumination, view-point alignment, and deterministic extractor settings. Mode B is therefore suited primarily to severely bandwidth-constrained deployments with calibrated observation geometry; Mode A (compact feature set, L_2 matching) is the default for general operation.

UWB NLOS. The $\beta=0.15$ per-hop noise growth assumes benign (line-of-sight) propagation. A fully cluttered (non-line-of-sight) deployment would require a measured-NLOS-aware $\beta(k)$; the current model is best interpreted as a rural/line-of-sight baseline, with NLOS calibration deferred to future work.

Temporal decay. The $\lambda = 10^{-3} \text{ s}^{-1}$ decay assumes quasi-static terrain. Active combat zones can alter descriptors on shorter timescales than 11.5 min, motivating a learned adaptive λ .

Re-admission. The re-admission protocol for previously excluded UAVs was not evaluated. In adversarial settings, an attacker with sustained physical access could repeatedly re-enrol compromised vehicles; mitigations include out-of-band operator authorisation and increasing T_{warmup} with re-admission count.

B. Deployment Considerations

Implementation targets commercial off-the-shelf embedded platforms capable of approximately 50–100 TOPS of inference throughput, standard short-range UWB ranging radios, MEMS- or tactical-grade inertial sensors, and a secure element for Ed25519 key storage. The navigation module integrates without modification of the underlying flight-control firmware (PX4/ArduPilot via MAVLink or proprietary autopilots via Ethernet). Because the follower tier requires no camera, imagery database, or GPU-class accelerator, a large swarm can combine a small number of fully-equipped anchor UAVs with many minimally-equipped followers, yielding a substantial

TABLE XIV: Design-space coverage of the proposed architecture vs. the closest published systems. Pillars: BFT consensus over navigation data, anchor/follower hierarchy, persistent cross-round reputation, automatic anchor-loss failover, GNSS-spoofing detection, multi-hop position relay. The right-most columns capture the architectural choices that gate compatibility with flight-targeted control. The proposed system’s pillars are design intent; their empirical realisation is the purpose of the experiments in Section VI. Note on BFTL’s “reputation” pillar credit: in the BFTL formalism [14] this entry refers to blockchain-weighted route commitments rather than reputation-weighted noise-covariance modulation of the state estimator (the mechanism in our architecture); the two are distinct technical commitments under the shared “reputation” label.

Pillar / Capability	Ours	SwarmRaft	BDGD	BFTL
BFT consensus	✓	(~)	✓	✓
Anchor/follower hierarchy	✓	–	–	✓
Persistent reputation	✓	–	✓	✓
Automatic failover	✓	(~)	–	–
GNSS-spoofing detection	✓	✓	✓	–
Multi-hop position relay	✓	–	–	✓
Continuous-state est.	yes	yes	yes	no
Hardware UAVs tested	no	no	no	no

reduction in per-UAV navigation hardware cost at the swarm level.⁶

C. Design-Space Comparison with Prior Art

Table XIV summarises the design-space comparison against the closest published systems, including BFTL [14] as the four-pillar prior-art benchmark. The comparison is structured by which of the six target pillars each system supports, and on the qualitative architectural choices (continuous-state estimator vs. discretised maps; hardware tested vs. simulation-only). Per-metric quantitative entries for the proposed system are deferred to the empirical-validation phase (Section VI); the analytical predictions of Section V establish the target numerical ranges. The proposed architecture is, to our knowledge, the only published design that simultaneously addresses all six pillars within a continuous-state flight-targeted estimator. BFTL is the closest prior art on pillar coverage (BFT, hierarchy, reputation, relay); it commits routes to a blockchain (Section II) and operates over discretised graph-structured environments. The present architecture adds automatic failover and GNSS-spoofing detection while replacing discretised map commitments with an incremental factor-graph estimator, which is the substantive gap BFTL leaves open.

D. Future Work

Four directions stand out. *First*, hardware validation at 10–20 UAV scale with injected adversarial behaviour, which our systematic review suggests has not yet been reported for BFT-aware UAV navigation. The concurrent Swarm Oracle work [56] demonstrates such validation on 12 ground

⁶A representative bill-of-materials comparison between the anchor and follower profiles is part of the associated patent application and is not reproduced here; a quantitative cost-multiplier figure is deferred to a future deployment-engineering paper where component-level pricing can be reported transparently.

TABLE XV: Reproducibility summary for the three abstract Monte Carlo simulators (Sections VIII–X). All under <https://github.com/ok-research/bft-uav-swarm-validation> tag v1.1; dependencies are numpy and matplotlib; run `python run_experiment.py` from the named sub-directory.

Sub-directory	Lines	Wall-clock	Seeds
<code>asymmetric_decay/</code>	≈ 140	10–15 min	{0, 1, 2}
<code>clamp_convergence/</code>	≈ 165	5–8 min	{0, 1, 2}
<code>byzantine_scaling/</code>	≈ 100	3–5 min	{0, 1, 2, 3, 4}

robots for blockchain-oracle scenarios; extending this validation methodology to GNSS-denied UAV flight is a natural next step. *Second*, a head-to-head comparison against the original SwarmRaft [12] codebase. *Third*, adversarial multi-hop error characterisation—closing a gap flagged explicitly in the BDGD paper [13]. *Fourth*, learned role adaptation using reinforcement-learning schemes such as CI-HRL [53] or semantic coordination as in RALLY [52], integrated with the Byzantine-resilient substrate presented here.

E. Code and Data Availability

The complete validation lab—ROS 2 nodes, sweep configurations, analysis scripts, Docker recipes, and raw production-sweep CSVs for every measurement in Section VI—is released at <https://github.com/ok-research/bft-uav-swarm-validation-lab> under the MIT licence; a tagged release matching the present manuscript is archived on Zenodo with DOI 10.5281/zenodo.20402935. The repository’s `docs/results/phase-{1..8}.md` files document each phase, including reproducibility pins (Docker image digest, git commit at sweep start, manifest of sweep parameters). A paper-claim to data-file mapping is provided as `CODE_TO_CLAIM_MAPPING.md` at the repository root. Sentinel-2 imagery [70] used for Terrain-Referenced Navigation is publicly available through the Copernicus Open Access Hub (<https://scihub.copernicus.eu/>) under the Copernicus Data Space open-access licence; no custom imagery dataset is generated by the present code release.

The three abstract Monte Carlo simulators of Sections VIII–X were released with an earlier version of this manuscript at <https://github.com/ok-research/bft-uav-swarm-validation> (tag v1.1); the present release supersedes them on the integrated-system axis while retaining them as standalone abstract-MC instruments. Per-experiment details (sub-directory, line count, wall-clock, explicit seeds) are summarised in Table XV.

XII. CONCLUSION

We presented and empirically validated a Byzantine-fault-tolerant hierarchical navigation architecture for GNSS-denied UAV swarms that integrates, within a continuous-state flight-targeted factor-graph estimator, six previously disjoint capabilities: BFT consensus, dynamic anchor/follower hierarchy, reputation-weighted multi-hop position relay, sub-10-second automatic failover, UWB-based GNSS-spoofing detection, and load balancing under progressive attrition. A systematic review of approximately 90 publications from 2020–2026 indicates

that, to our knowledge, no prior published system has combined more than four of these pillars, and none has done so in a continuous-state flight-targeted estimator. The “six pillars” frame is the organising choice of the present paper; alternative axes (e.g., treating load balancing as a separate concern rather than a pillar, or splitting failover into detection and reassignment sub-pillars) would yield similar gap analyses for the literature surveyed. The central primitive, visual Proof of Observation, ties trust to terrain-anchored evidence (physically bound to terrain geometry, illumination, and the learned extractor; not forgeable by a vehicle that has not observed the location under the threat model adopted here) and integrates naturally with factor-graph state estimation via asymmetric reputation-weighted noise. Theoretical analysis exposes an explicit trade-off between quorum size and scalable Byzantine tolerance, gives a closed-form parametric estimate of the naive-Byzantine exclusion time ($T_{\text{exclusion}} \approx 215$ s as a loose upper budget under the most conservative parameter substitutions; the measured median exclusion time is 5–15 s under realistic operational parameters), and quantifies multi-hop error propagation. The default $M=3$ configuration safely excludes $f \leq 2$ absolute Byzantine peers under the Proposition 1 safety regime; detection of larger Byzantine fractions (Section X) does not extend the safe-exclusion regime, which would require scaling M at the cost of slower exclusion. Per the rule of three, the 0/30 false-exclusion result of Phase 2 is statistically consistent with a true rate up to $\approx 10\%$; an expanded Phase 2 corpus would narrow this bound and is identified as follow-up.

Two findings from the empirical campaign (Section VI) should be retained beyond the headline metrics tabulated in the Abstract and Section VI. First, the empirical per-hop GDOP multiplier $\delta = 0.088$ measured at three relay hops is roughly half of the conservative $\delta = 0.15$ assumed in Section V-C; the exponential family is confirmed, but the design margin against the 100-m operational ISR CEP threshold is wider than originally claimed, leaving headroom for the deeper relay chains needed at $N \gtrsim 50$. Second, the Section VII side-by-side against a vanilla-Raft leader-election protocol isolates the cost of Byzantine resilience at approximately one second of failover overhead: vanilla-Raft is compromised indefinitely under a single “leader-claimer” attacker, while the proposed system excludes the analogous Byzantine anchor within the Phase 5 detection envelope. The two qualitatively distinct outcomes (rather than the absolute timings) are what crash-fault practitioners need when sizing the Byzantine premium.

The work as published is bounded by two practical limits. The Tier-B Python harness used for Phases 5 and 6 places UAVs at fixed positions, suppressing trajectory divergence and IMU dead-reckoning drift; closing the gap between Tier-B protocol-layer validation and full Gazebo flight dynamics is the dominant near-term work item. Concrete follow-up directions are listed in Section XI-D.

AI Disclosure

All software released under the companion repository (Section XI-E)—ROS 2 nodes, sweep configurations, analysis

scripts, Docker recipes—together with the abstract Monte Carlo simulators released with an earlier version of this manuscript, was developed with LLM-assisted code generation (Anthropic Claude). All numerical results reported in Section VI and Section VII were obtained by deterministic re-execution of the released code; each production sweep is reproducible from the corresponding `manifest.json`. The manuscript text was drafted by the author; LLM assistance was used to triage early framing options and check internal consistency across sections. Every numerical claim is sourced from a named workspace artefact.

Competing Interests

The author has filed two pending patent applications at the DPMA (Aktenzeichen 10 2026 001 712.2 and a related application covering Sections IV-B through IV-G of the present manuscript). No other competing interests are declared.

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This work received no external funding. Compute resources used for the empirical validation campaign were obtained from the author’s personal vast.ai account at negligible compute cost (Phases 1 and 2, GPU-bound Proof-of-Observation pipeline); Phases 3–8 ran on the author’s local workstation at zero marginal cost.

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Oleksandr Kalynovskiy is an independent researcher. His research interests include Byzantine fault tolerance, resilient cooperative navigation, and GNSS-denied localization for autonomous UAV swarms.